

2. Digital Logic and Microprocessor

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Syllabus

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

2.6 Interrupt operations: Interrupt, Interrupt Service Routine, and Interrupt Processing. (AExE0206)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

MICROPROCESSOR SYSTEM

- A microcomputer consists of a set of components or modules of three basic types CPU memory and I/O units which communicate with each other.

PIN CONFIGURATION OF 8085

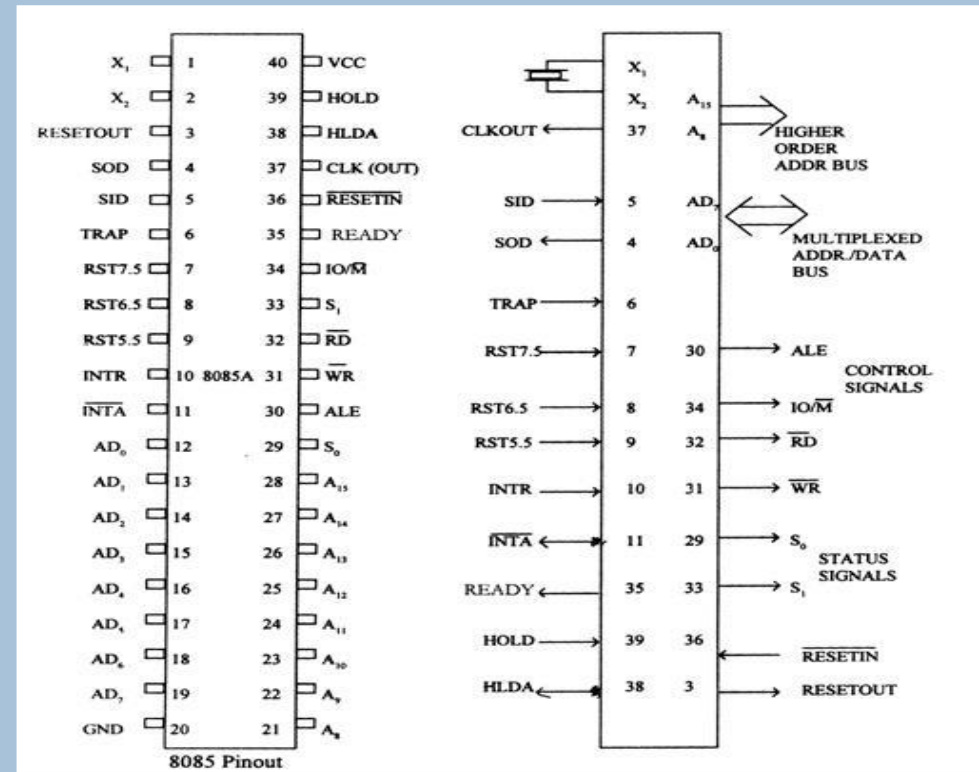


Fig (a) - Pin Diagram of 8085 & Fig(b) - logical schematic of Pin diagram

PIN CONFIGURATION OF 8085

- The microprocessor is a clock-driven semiconductor device consisting of electronic logic circuits manufactured by using either a large-scale integration (LSI) or very-large-scale integration (VLSI) technique.
- The microprocessor is capable of performing various computing functions and making decisions to change the sequence of program execution.
- In large computers, a CPU implemented on one or more circuit boards performs these computing functions.
- The microprocessor is in many ways similar to the CPU, but includes the logic circuitry, including the control unit, on one chip.

PIN CONFIGURATION OF 8085

- The microprocessor can be divided into three segments for the sake clarity, arithmetic/logic unit (ALU), register array, and control unit.
- 8085 is a 40 pin IC, DIP package. The signals from the pins can be grouped as follows
 - 1. Power supply and clock signals
 - 2. Address bus
 - 3. Data bus
 - 4. Control and status signals
 - 5. Interrupts and externally initiated signals
 - 6. Serial I/O ports

PIN CONFIGURATION OF 8085

Power supply and Clock frequency signals:

- **Vcc** : + 5 volt power supply
- **Vss** : Ground
- **X1, X2** : Crystal or R/C network or LC network connections to set the frequency of internal clock generator.
- The frequency is internally divided by two. Since the basic operating timing frequency is 3 MHz, a 6 MHz crystal is connected externally.
- **CLK (output)** :Clock Output is used as the system clock for peripheral and devices interfaced with the microprocessor.

PIN CONFIGURATION OF 8085

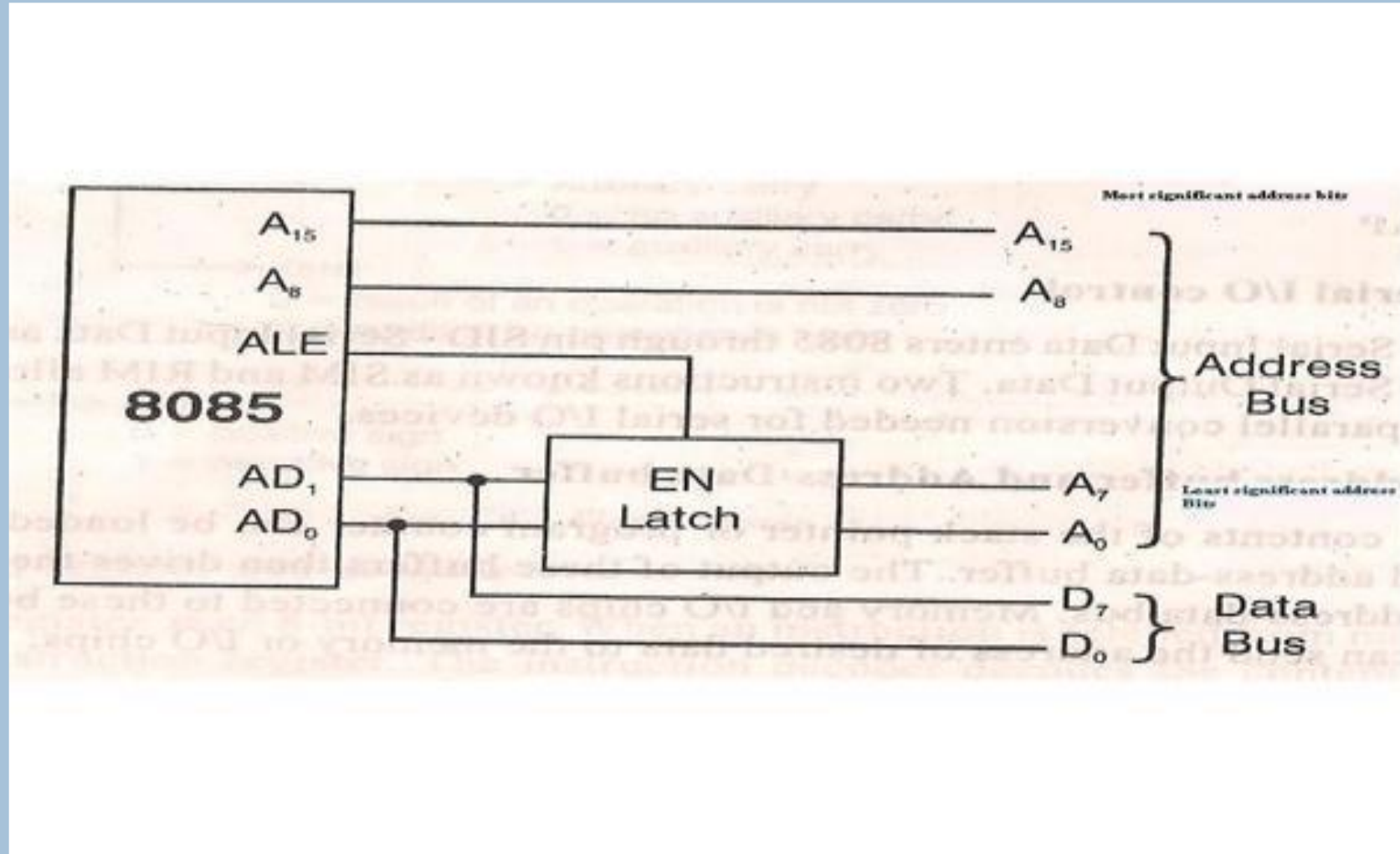
2. Address Bus:

- **A8 - A15**
- It carries the most significant 8 bits of the memory address or the 8 bits of the I/O address.

3. Multiplexed Address / Data Bus:

- **AD0 - AD7**
- These multiplexed set of lines used to carry the lower order 8 bit address as well as data bus.
- During the opcode fetch operation, in the first clock cycle, the lines deliver the lower order address A0 - A7.
- In the subsequent IO / memory, read / write clock cycle the lines are used as data bus.
- The CPU may read or write out data through these lines.

PIN CONFIGURATION OF 8085



PIN CONFIGURATION OF 8085

Control and Status signals:

- These signals include two control signals (RD & WR) three status signals (IO/M, S1 and So) to identify the nature of the operation and one special signal (ALE) to indicate the beginning of the operations.
- **ALE (output)** - Address Latch Enable.
 - This signal helps to capture the lower order address presented on the multiplexed address / data bus. When it is the pulse, 8085 begins an operation. It generates AD0 - AD7 as the separate set of address lines A0 –A7.
- **RD (active low)** - Read memory or IO device.
 - This indicates that the selected memory location or I/O device is to be read and that the data bus is ready for accepting data from the memory or I/O device.

PIN CONFIGURATION OF 8085

- **WR (active low)** - Write memory or IO device.
 - This indicates that the data on the data bus is to be written into the selected memory location or I/O device.
- **IO/M' (output)** - Select memory or an IO device.
 - This status signal indicates that the read / write operation relates to whether the memory or I/O device.
 - It goes high to indicate an I/O operation.
 - It goes low for memory operations.

PIN CONFIGURATION OF 8085

Status Signals:

- It is used to know the type of current operation of the microprocessor.

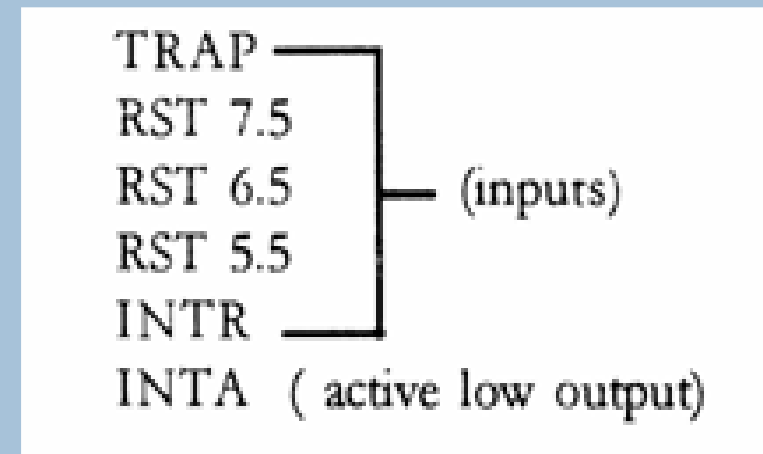
IO/M(Active Low)	S1	S2	Data Bus Status (Output)
0	0	0	Halt
0	0	1	Memory WRITE
0	1	0	Memory READ
1	0	1	IO WRITE
1	1	0	IO READ
0	1	1	Opcode fetch
1	1	1	Interrupt acknowledge

PIN CONFIGURATION OF 8085

Interrupts and Externally initiated operations:

➤ They are the signals initiated by an external device to request the microprocessor to do a particular task or work.

➤ There are five hardware interrupts called,



➤ On receipt of an interrupt, the microprocessor acknowledges the interrupt by the active low INTA (Interrupt Acknowledge) signal.

PIN CONFIGURATION OF 8085

➤ **Hold (Input)**

- This indicates peripheral controller requesting the bus.

➤ **HLDA (Output)**

- This indicates the acknowledgement for the Hold request.

➤ **READY (Input)**

- It is used to delay the microprocessor read and write cycles until a slow responding peripheral is ready to send or accept data.
- Memory and I/O devices will have slower response compared to microprocessors.
- Before completing the present job such a slow peripheral may not be able to handle further data or control signal from CPU.
- The processor sets the READY signal after completing the present job to access the data.
- The microprocessor enters into WAIT state while the READY pin is disabled.

PIN CONFIGURATION OF 8085

➤ **Reset In (input, active low)**

- This signal is used to reset the microprocessor.
- The program counter inside the microprocessor is set to zero.
- The buses are tri-stated.

➤ **Reset Out (Output)**

- It indicates CPU is being reset.
- Used to reset all the connected devices when the microprocessor is reset.

Single Bit Serial I/O ports:

- **SID (input)** - Serial input data line
- **SOD (output)** - Serial output data line
- These signals are used for serial communication.

Bus structure

- A microcomputer consists of a set of components or modules of three basic types CPU memory and I/O units which communicate with each other.
- A bus is a communication pathway between two or more such components.
- A bus actually consists of multiple communication pathway or lines. Each line is capable of transmitting signals representing binary 1 and 0.
- Several lines of the bus can be used to transmit binary data simultaneously.
- The bus that connects major microcomputer components such as CPU, memory or I/O is called the system bus.
- System bus consists of number of separate lines. Each line assigned a particular function.
- Fundamentally in any system bus the lines can be classified into three group buses.

Bus structure

Data Bus:

- Data bus provides the path for monitoring data between the system modules. The bus has various numbers of separate lines like 8, 16, 32, or 64. Which referred as the width of data bus .These number represents the no. of bits they can carry because each carry 1 bit.

Address Bus:

- Each Lines of address bus are used to designate the source or destination of the data on data bus.
- For example, if the CPU requires reading a word (8, 16, 32) bits of data from memory, it puts the address of desired word on address bus. The address bus is also used to address I/O ports. Bus width determines the total memory the up can handle.

Bus structure

Control Bus:

- The control bus is a group of lines used to control the access to control signals and the use of the data and address bus. The control signals transmit both command and timing information between the system modules. The timing signals indicate the validity of data and address information, where as command signals specify operations to be performed. Some of the control signals are:
- **Memory Write (MEMW):** It causes data on the bus to be loaded in to the address location.
- **Memory Read (MEMR):** It causes data from the addressed location to be placed on the data bus.
- **I/O Write (IOW):** It causes the data on the bus to be output to the addressed I/O port.
- **I/O Read (IOR):** It causes the data from the addressed I/O port to be placed on the bus.
- **Transfer Acknowledge:** This signal indicates that data have been accepted from or placed on the bus.

Bus structure

- **Bus Request:** It is used to indicate that a module wants to gain control of the bus.
- **Bus Grant:** It indicates that a requesting module has been granted for the control of bus.
- **Interrupt Request:** It indicates that an interrupt has been pending.
- **Interrupt Acknowledge:** it indicates that the pending interrupt has been recognized.

Bus types

Synchronous Bus:

- In a synchronous bus, The Occurrence of the events on the bus is determined by a clock .
- The clock transmits a regular sequence of 0's & 1's of equal duration. All the events start at beginning of the clock cycle.

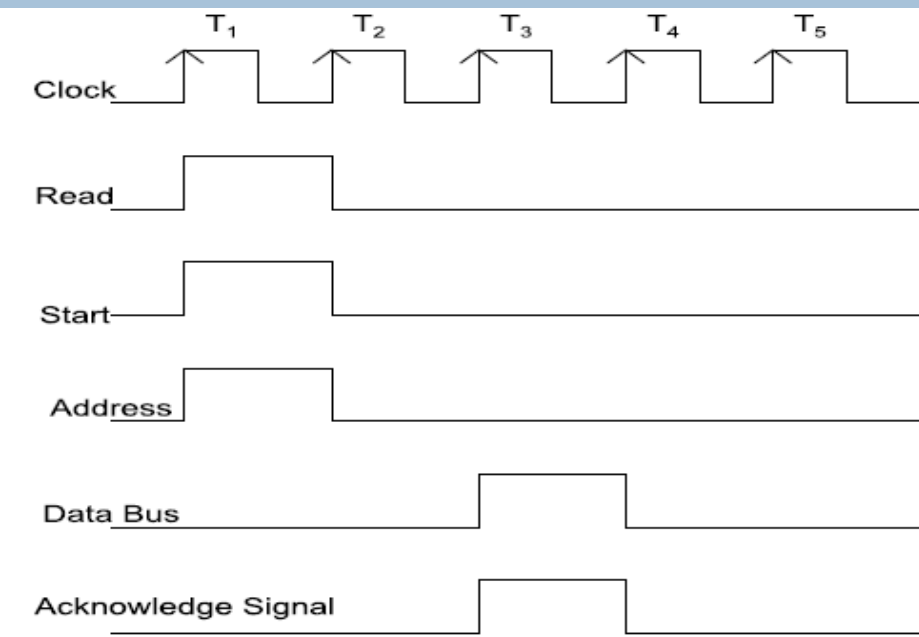


Fig: Synchronous Read Operation

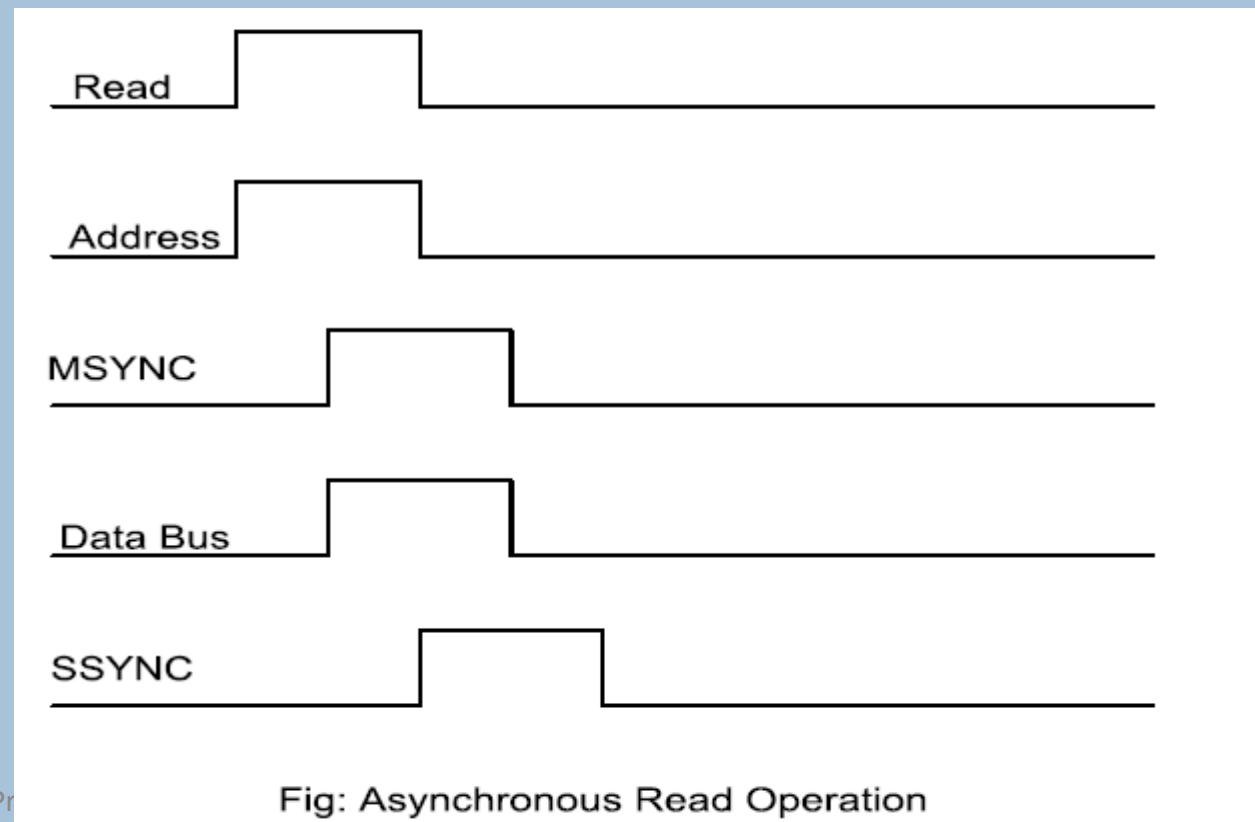
Bus types

- Here the CPU issues a START signal to indicate the presence of address and control information on the bus.
- Then it issues the memory read signal and places the memory address on the address bus.
- The addressed memory module recognizes the address and after a delay of one clock cycle it places the data and acknowledgment signal on the buses.
- In synchronous bus, all devices are tied to a fixed rate, and hence the system can not take advantage of device performance but it is easy to implement.

Bus types

Asynchronous Bus:

- In an asynchronous bus, the timing is maintained in such way that occurrence of one event on the bus follows and depends on the occurrence of previous event.

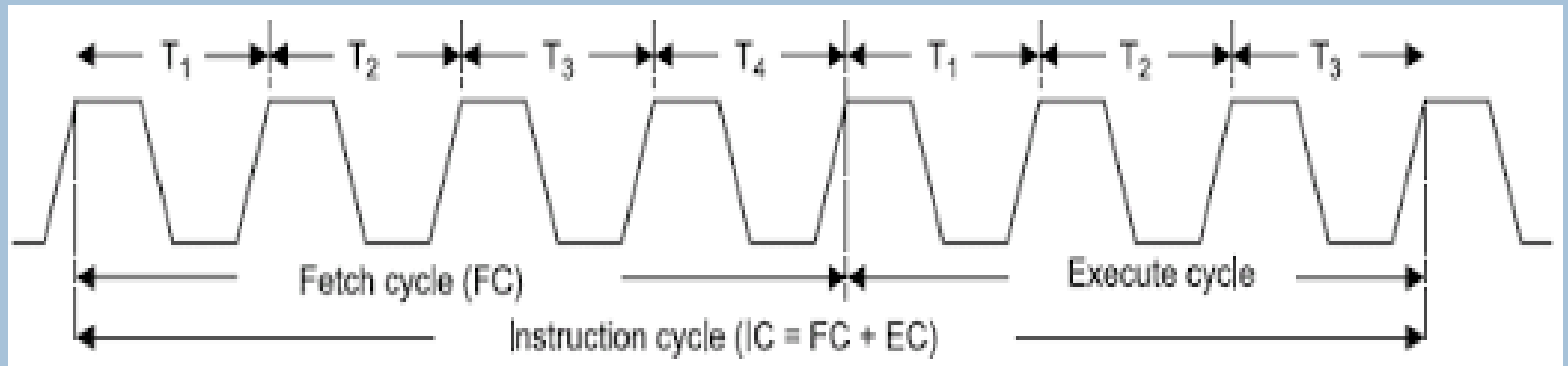


Bus types

- Here the CPU places Memory Read (Control) and address signals on the bus.
- Then it issues master synchronous signal (MSYNC) to indicate the presence of valid address and control signals on the bus.
- The addressed memory module responds with the data and the slave synchronous signal (SSYNC)

Timing diagram

- It is a graphical representation. It represents the execution time taken by each instruction in a graphical format. The execution time is represented in T-state.



Instruction cycle

- It is defined as the time required to complete the execution of an instruction.
- The necessary steps that the CPU carries out to fetch an instruction and necessary data from the memory and to execute it constitute an instruction cycle.
- An instruction cycle consists of fetch cycle and execute cycle.
- In fetch cycle CPU fetches op-code from the memory.
- The necessary steps which are carried out to get data if any from the memory and to perform the specific operation specified in instruction constitute an execute cycle.
- The total time required to execute an instruction is given by $IC = FC + EC$
- The 8085 consists of 1-5 machine cycle or operation.

fetch cycle

- The first byte of an instruction is its op-code.
- The content of the program counter, which is the address of the memory location where op-code is available, is send to the memory.
- The memory places the op-code on the data bus so as to transfer it to CPU.
- The entire process takes 3 clock cycle and then the instruction is decoded in next one clock cycle.

Execute cycle

- The op-code from the memory goes to the IR, from the IR it goes to the decoder which decodes instruction. After the instruction is decoded execution begins.
- If the operand is in general purpose register, execution is performed immediately.
- If an instruction contains data or operand address, then CPU has to perform some read operations to get the desired data.
- In some instruction write operation is performed. In write cycle data are sent from the CPU to the memory of an o/p device.
- In some cases execute cycle may involve one or more read or write cycle or both.

machine cycle

- It is defined as the time required to complete one operation of accessing memory i/p, o/p or acknowledging the external request. This cycle may consists of 3 to 6 T states.
- Op-code Fetch Cycle
- Memory Read Cycle (3T)
- Memory Write Cycle (3T)
- I/O Read Cycle (3T)
- I/O Write Cycle (3T)
- Interrupt acknowledge
- Bus idle

T-state:

- It is defined as one sub-division of the operation performed in one clock period. These sub-division are internal states synchronized with system clock and each T state precisely equal to one clock period.

Opcode fetch machine cycle

- The first machine cycle of every instruction is opcode fetch cycle in which the 8085 finds the nature of the instruction to be executed.
- In this machine cycle, the microprocessor places the contents of PC on the address bus then by reading operation it reads the op-cod of an instruction from determined memory location. The length of this cycle is not fixed.

Step1: (T1 state)

- The 8085 processor places the contents of program counter on the address bus, activate the ALE and send the status signals IO/M', S1, and S0 with logical status (0 1 1) respectively.

Step 2: (T2 state)

- The low order address disappears from AD0-AD7 lines. Also, 8085 processor activates the RD signals to enable the addressed memory location which places its contents on the data bus (AD0-AD7).

Opcode fetch machine cycle

Step 3: (T3 state)

- The processor loads the contents of data bus on its Instruction Register and deactivates the RD signal to disable the memory devices.

Step4: (T4 state)

- Microprocessor decodes the instruction and performed the task specified into instruction.

Step5: (T5 & T6 states)

- The processor performs stack write, internal 16 bits, or conditional return operations depending upon the type of instruction.
- One byte instructions those operate on 16 bit data are executed in T5 & T6. For example DCX H, PCHL, SPHL, INX H, etc.

Opcode fetch machine cycle

Instruction that takes 6-T for opcode fetch

CRISP

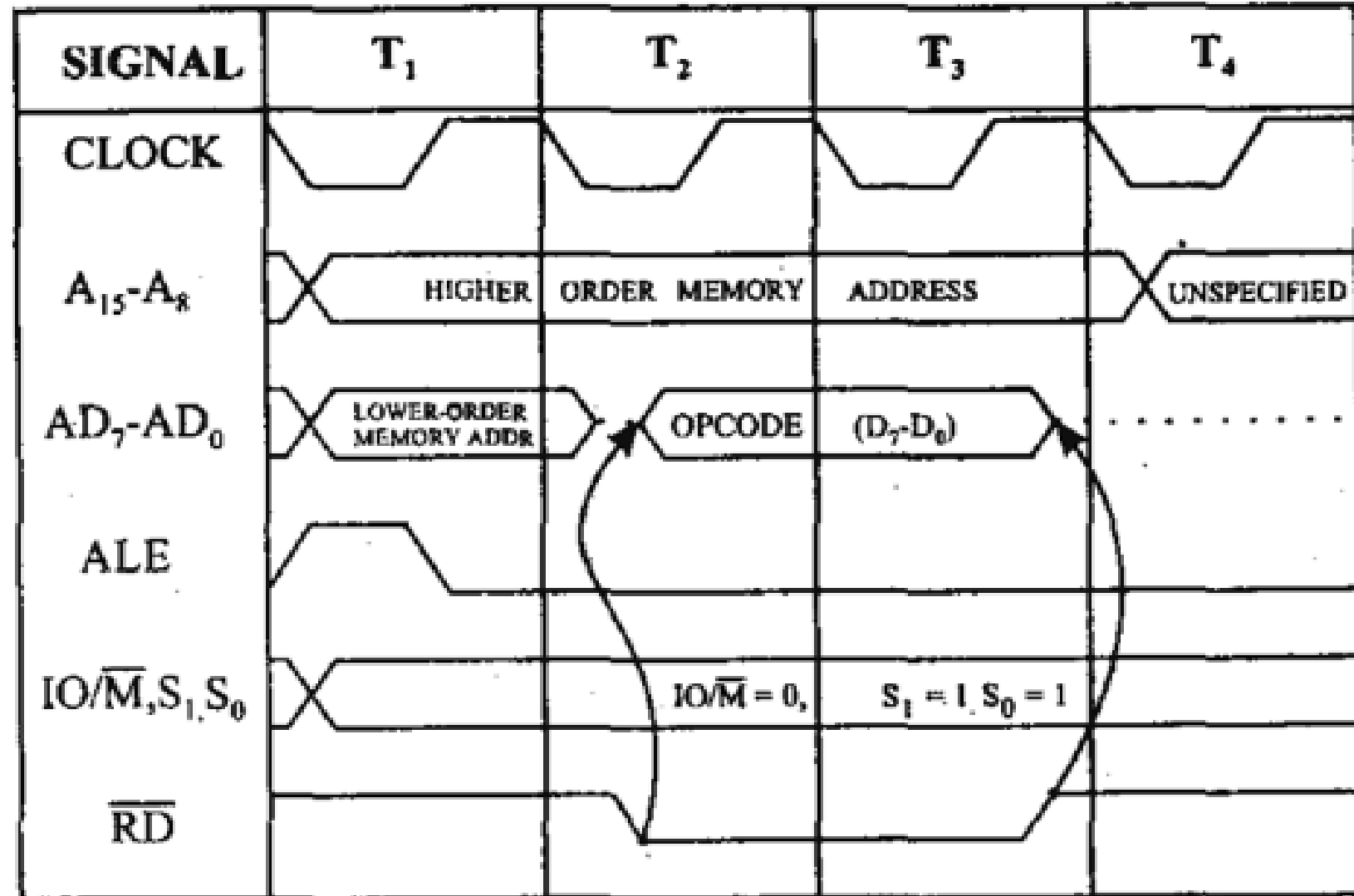
C- CALL, Conditional
RET.

R- RST

I- INX ,DCX

S- SPHL, PCHL

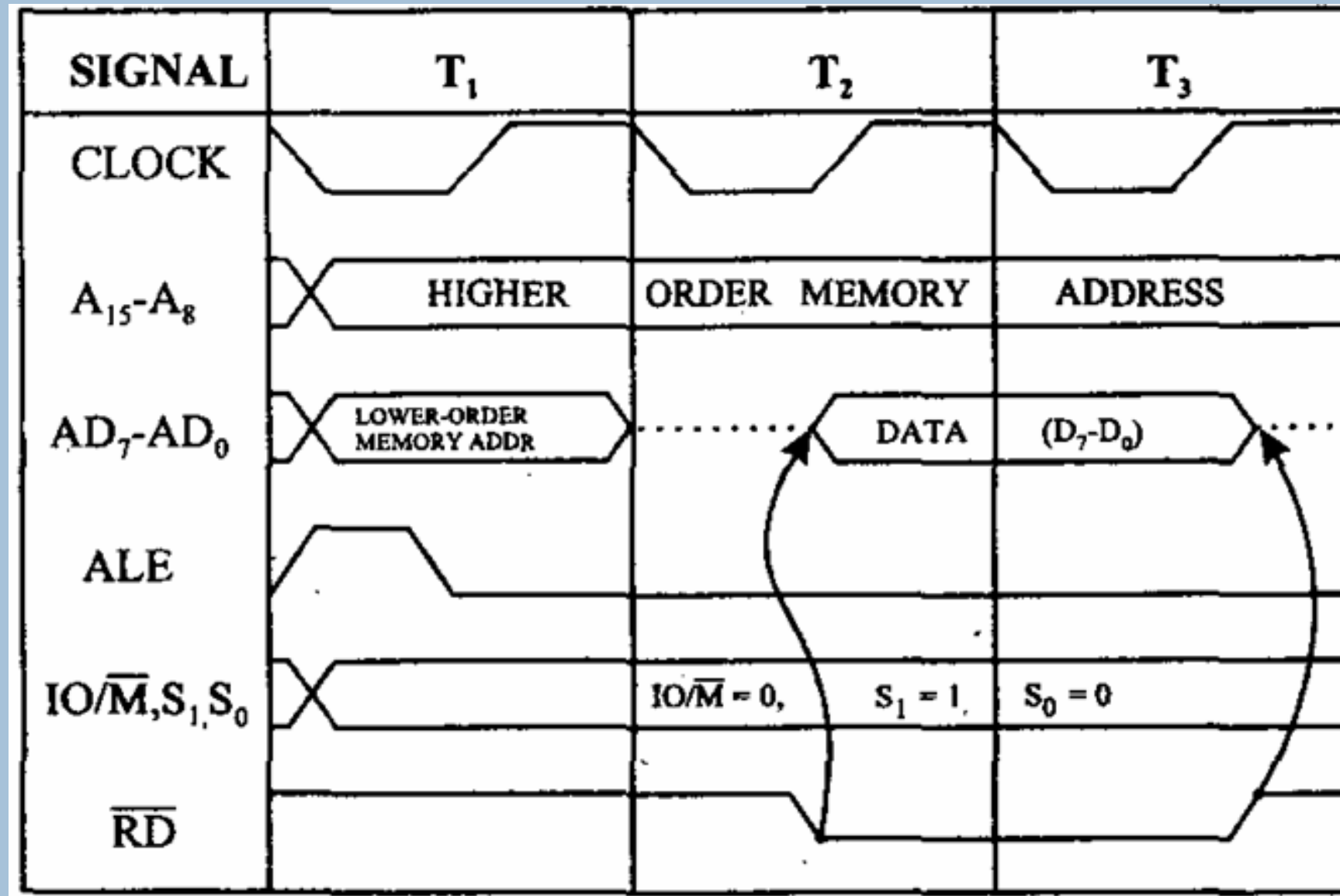
P- PUSH



MEMORY READ machine cycle

- The microprocessor executes the memory read cycle to read the data from RAM or ROM memory. 8085 processor executes this machine cycle in 3 T-states. Steps below show the details of this machine cycle:
- **Step1 (T1 state):**
 - processor places the address on the address lines from SP, Rp, or PC and activates ALE in order to latch low-order of address. Also, it sends the status signals IO/M', S1, and S0 with logical status (0 1 0) for memory read machine cycle.
- **Step2 (T2 state):**
 - 8085 processor activates the RD' signals to enable the addressed memory location which places its contents on the data bus (AD0-AD7).
- **Step 3: (T3 state)**
 - The processor loads the contents of data bus on specified register (F, A, B, C, D, E, H, and L) and deactivates the RD' signal to disables the memory devices.

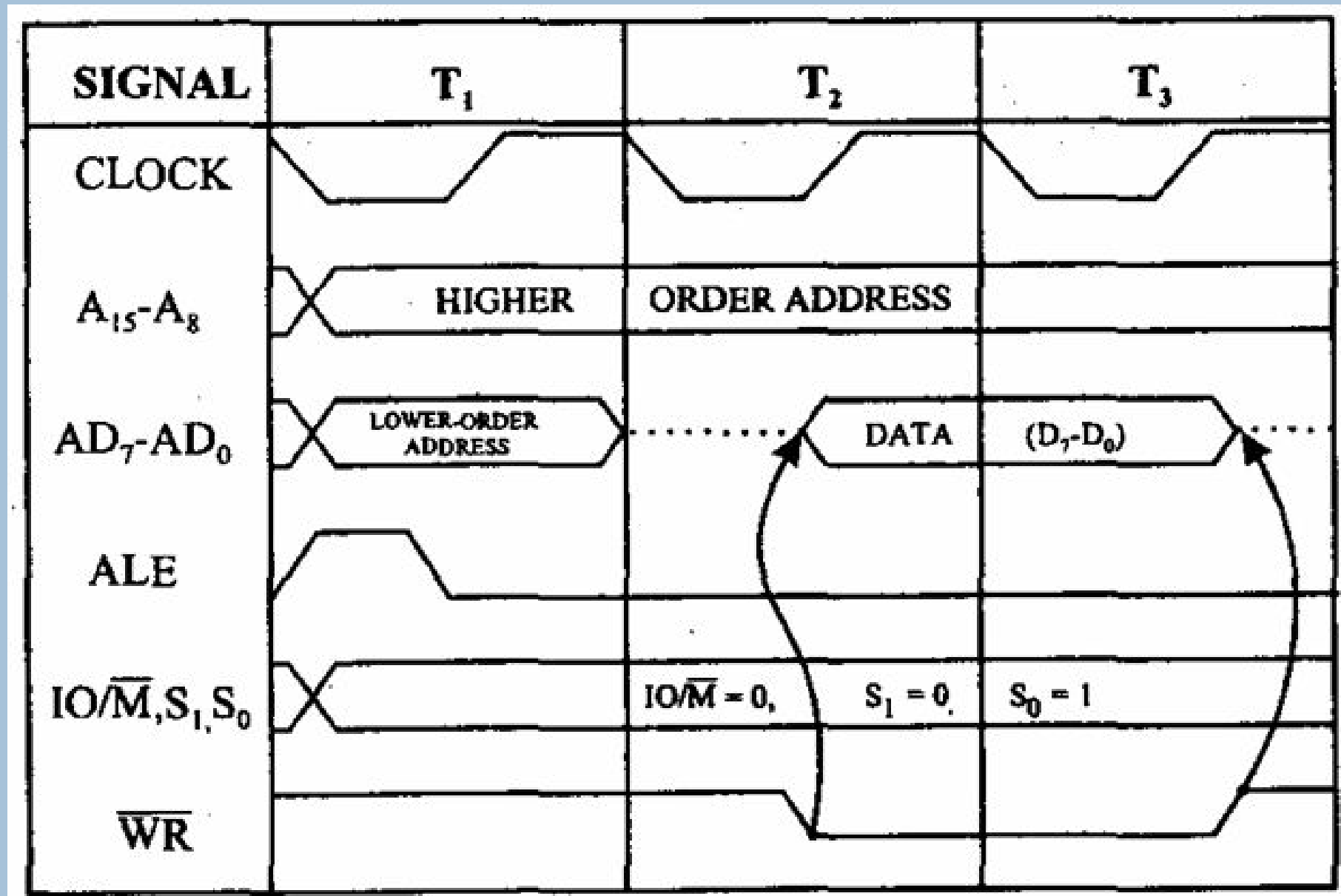
MEMORY READ machine cycle



MEMORY WRITE machine cycle

- The microprocessor executes the memory write cycle to store the data into RAM or stack memory. 8085 processor executes this machine cycle in 3 T-states. Steps below show the details of this machine cycle:
- **Step1 (T1 state):**
 - processor places the address on the address lines from SP or Rp and activates ALE in order to latch low-order of address. Also, it sends the status signals IO/M', S1, and S0 with logical status (0 0 1) for memory write machine cycle.
- **Step2 (T2 state):**
 - 8085 processor places the data on data bus and activates the WR' signal to writing data into addressed memory location.
- **Step 3: (T3 state)**
 - The processor deactivates the WR' signal which disables the memory device and terminates the write operation.

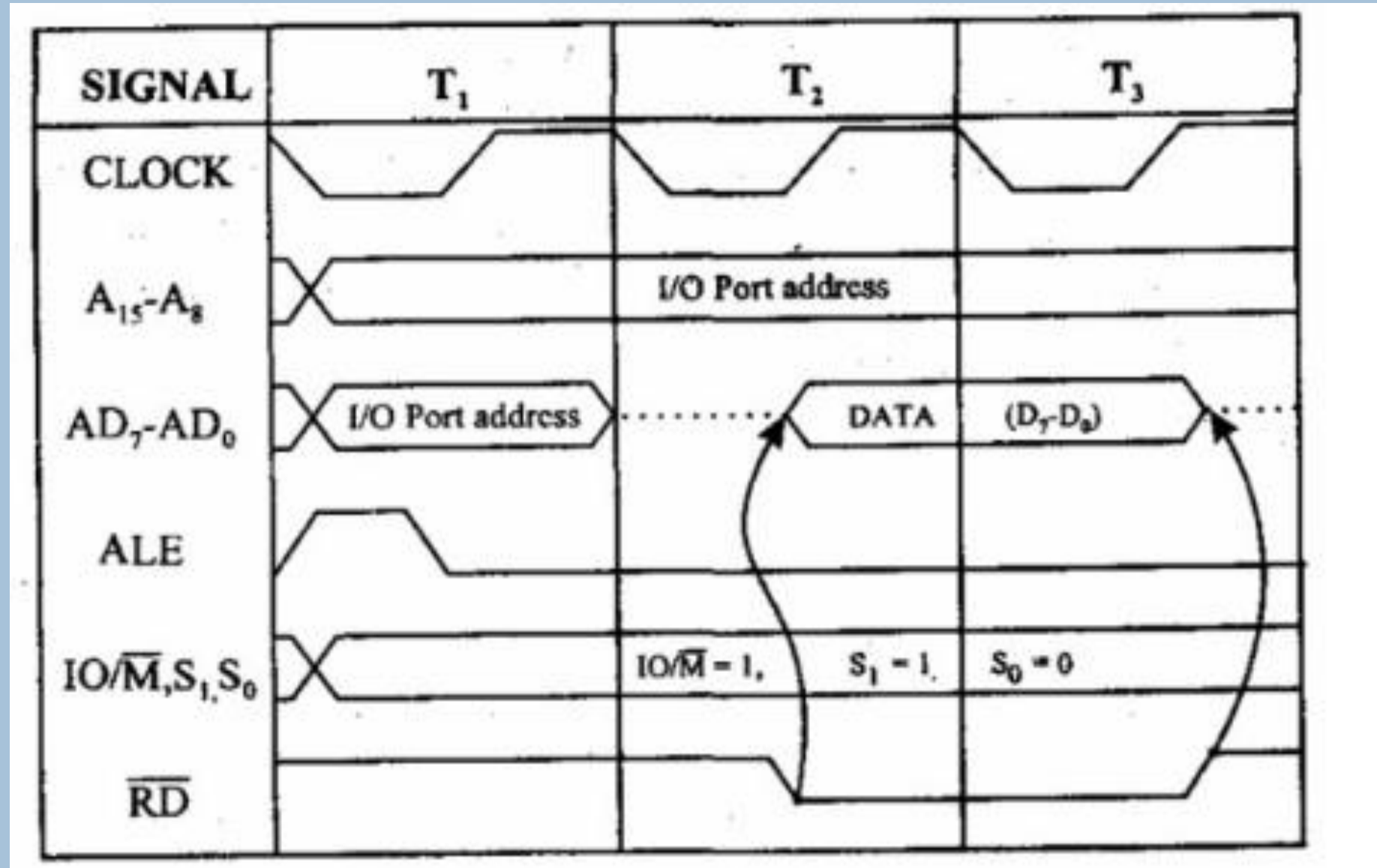
MEMORY WRITE machine cycle



IO READ machine cycle

- The microprocessor executes the IO read cycle to read the data from input device. 8085 processor executes this machine cycle in 3 T-states. Steps below show the details of this machine cycle:
- **Step1 (T1 state):**
 - processor places the address on the address lines from SP, Rp, or PC and activates ALE in order to latch low-order of address. Also, it sends the status signals IO/M', S1, and S0 with logical status (1 1 0) for IO read machine cycle.
- **Step2 (T2 state):**
 - 8085 processor activates the RD' signals to enable the addressed input device which places its contents on the data bus (AD0-AD7).
- **Step 3: (T3 state)**
 - The processor loads the contents of data bus on specified register (F, A, B, C, D, E, H, and L) and deactivates the RD' signal to disables the input device.

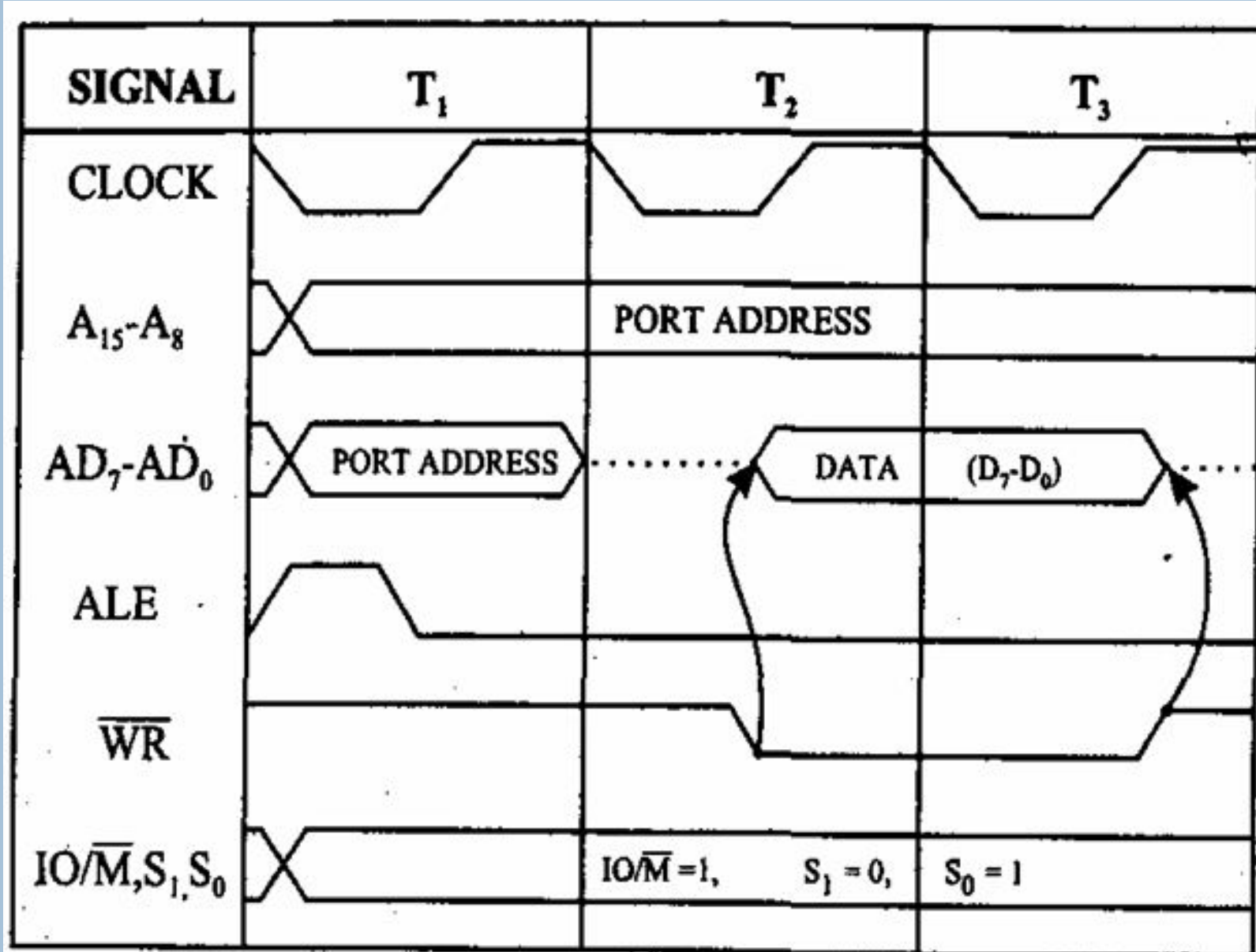
IO READ machine cycle



IO write machine cycle

- The microprocessor executes the IO write cycle to store the data into output device. 8085 processor executes this machine cycle in 3 T-states. Steps below show the details of this machine cycle:
- **Step1 (T1 state):**
 - processor places the address on the address lines from SP or Rp and activates ALE in order to latch low-order of address. Also, it sends the status signals with logical status (1 0 1) for IO write machine cycle.
- **Step2 (T2 state):**
 - 8085 processor places the data on data bus and activates the WR' signal to writing data into addressed output device.
- **Step 3: (T3 state)**
 - The processor deactivates the WR' signal which disables the output device and terminates the writing operation.

IO write machine cycle



memory

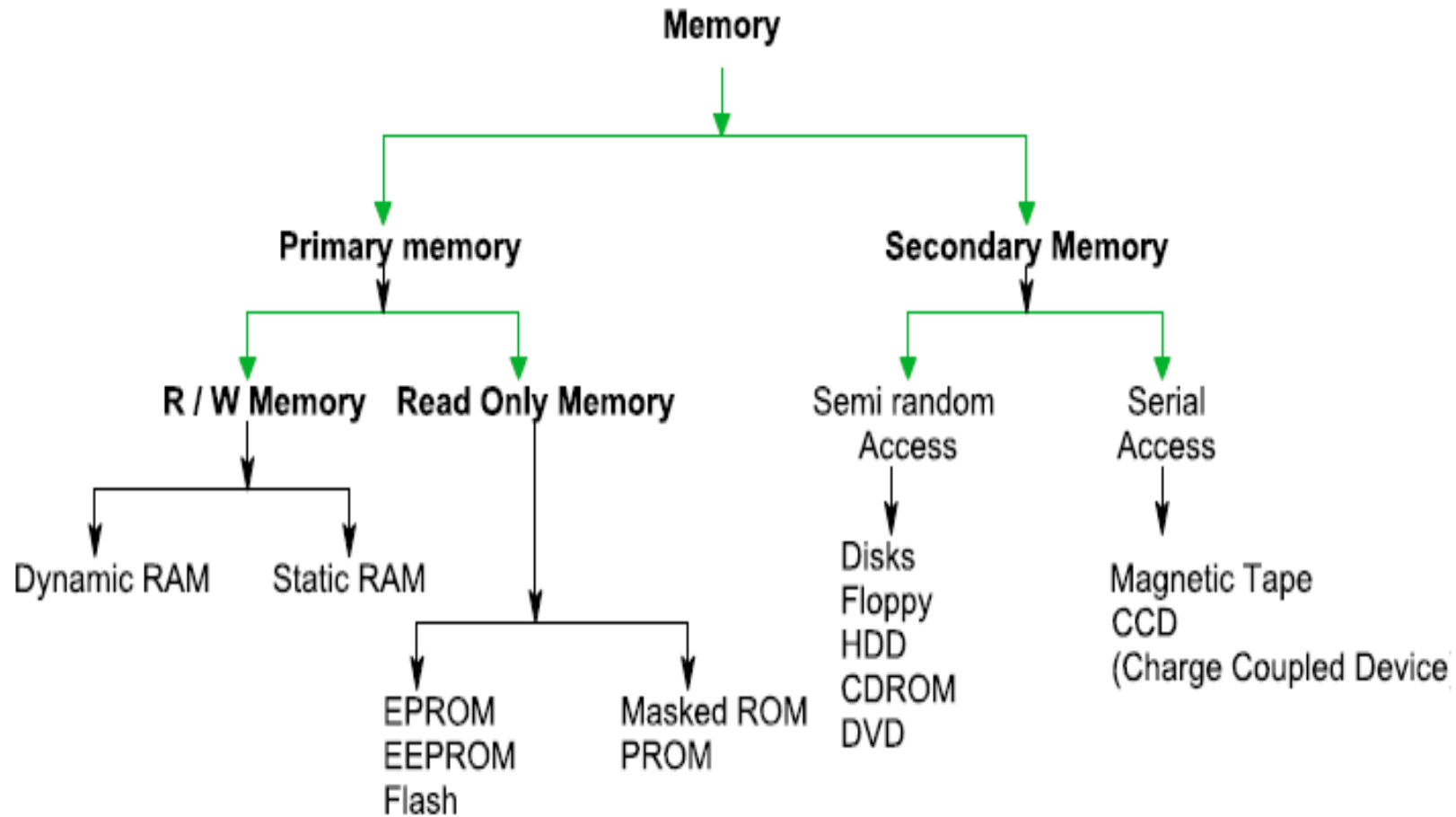


Fig: Classification of memory system

memory

- Memory is an essential component of the microcomputer system.
- It is used to store both instructions and data.
- Memory is made up of registers and the number of bits stored in a register is called memory word .
- Memory word is identified by an address .
- If microprocessor uses 16 bit address , then there will be maximum of $2^{16} = 65536$ memory addresses ranging from 0000H to FFFFH.
- There are various types of memory which can be classified in to two main groups i.e. Primary memory and Secondary memory.

memory

Processor Memory

- Processor memory refers to a set of microprocessor registers.
- They are used to hold temporary results when a computation is in progress Although use of such registers enhances the execution speed
- The cost involved in the approach forces a microcomputer designer to include only a few registers include the processor.
- In 8085 we have registers like A, B, C, D, E, H, L, SP, PC etc. to store data temporarily

Primary Memory

- It is the storage area where all programs are executed. The microprocessor can directly access only those items that are stored in the primary memory
- Hence, all programs and data must be within the primary memory prior to execution.
- Usually, the size of the primary memory is much larger than that of processor memory and its operating speed is much slower than processor's registers.

memory

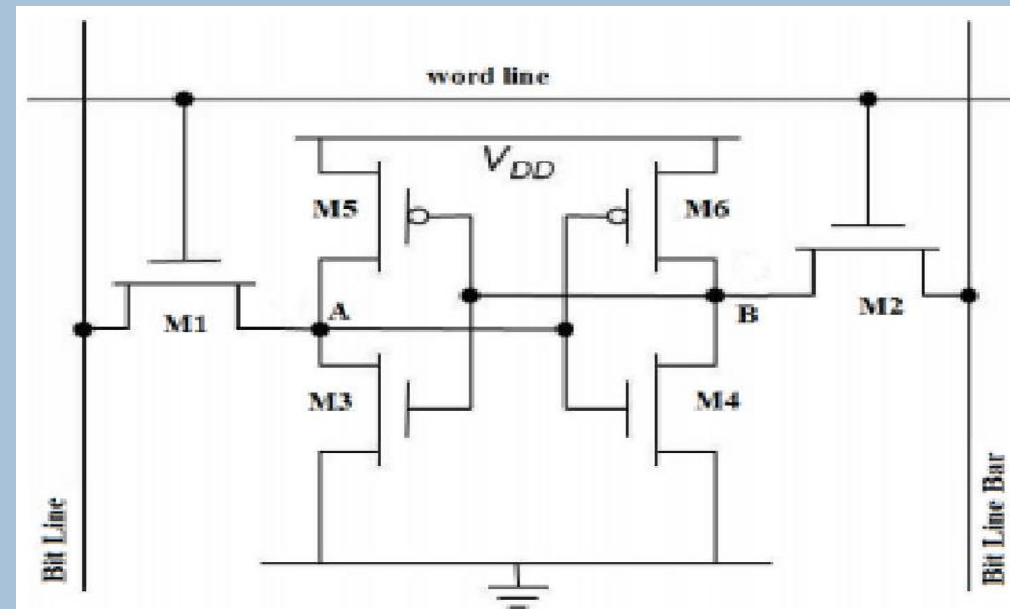
- Primary memories can be divided into two main groups
 - Read only memory (ROM)
 - Random Access memory .(RAM)

RAM

- It is used primarily for information that is likely to be altered, such as writing programs or receiving data.
- This memory is volatile Two types of RAM are available
- **Static RAM**
 - This memory is made up of flip flops and stores the bits as voltage.

memory

- Each memory cell requires six transistors.
- This memory is more expensive and power consuming than dynamic memory.
- It is called 'static' because the information doesn't need a constant update.
- These memories are commonly used for cache memory.
- This type of memory is very fast with access time is 15 to 30 nanoseconds but is physically bulky.

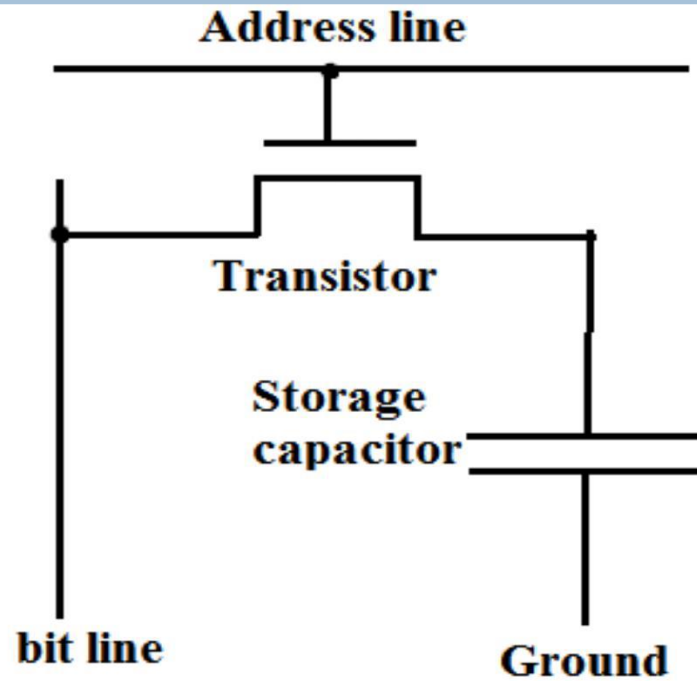


memory

Dynamic RAM (DRAM)

- Dynamic random access memory is an improvement over the expensive and bulky SRAM.
- DRAM uses a different approach to store data Information is stored as charges in a very small capacitor.
- If a charge exists in a capacitor, it's interpreted as 1 The absence of a charge will be interpreted as 0.
- Because DRAM uses capacitors there is a chance of leakage of charge.
- Thus it needs to use a constant refresh signal to keep the information in the memory (every few millisecond)
- DRAM technology allows several memory units, called cells to be packed at very high density.
- Therefore, these chips can hold very large amount of information.
- Most PCs today use DRAM.
- Access time for DRAM is 80 nanoseconds or more, slower than SRAM, and two or three times faster than ROM

memory



DRAM

<u>SRAM</u>	<u>DRAM</u>
1. SRAM has lower access time, so it is faster compared to DRAM.	1. DRAM has higher access time, so it is slower than SRAM.
2. SRAM is costlier than DRAM.	2. DRAM costs less compared to SRAM.
3. SRAM requires constant power supply, which means this type of memory consumes more power.	3. DRAM offers reduced power consumption, due to the fact that the information is stored in the capacitor.
4. Due to complex internal circuitry, less storage capacity is available compared to the same physical size of DRAM memory chip.	4. Due to the small internal circuitry in the one-bit memory cell of DRAM, the large storage capacity is available.
5. SRAM has low packaging density.	5. DRAM has high packaging density.

memory

Read only memory (ROM)

- ROM is a non volatile memory and can be read only.
- It is used to store data and programs that are not to be altered
- Among other things ROM is needed for storing an initial program called boot strap loader.
- The bootstrap loader is a program whose function is to start the computer software operating when power is turned on.
- Since RAM is volatile, its contents are destroyed when power is turned off . The contents of ROM remain unaltered after power is turned off and on again
- The startup of a computer consists of turning the power on and starting the execution of an initial program.
- Thus when power is turned on, the hardware of the computer sets the program counter to the first address of the bootstrap loader.
- The bootstrap program loads a portion of the operation of the operating system from disk to main memory and control is then transferred to the operating system, which prepares the computer for general use.

memory

Masked ROM

- They are permanent ROM recorded by masking Generally manufacturers use this process to produce ROM in large numbers

PROM

- These are unprogrammed ROM The fuses on the ROM are not burned.
- A programmer can program this ROM according to his needs.
- The information stored is permanent.

memory

EPROM

- These ROM can be reprogrammed and erased. Two types of such EPROM are available

UV EPROM

- The memory of such ROM can be erased by exposing the chip via a lid or window on the chip to ultraviolet light.
- The erase time generally varies between 10 to 30 minutes.
- The EPROM can be programmed by inserting the chip into a socket of the PROM programmer and providing proper addresses.
- The programming time varies from 1 to 2 minutes.

EEPROM

- This does not require UV rays to be erased It can be completely erased or have certain bytes changed, using electrical pulses
- Writing to EEPROM is slower than writing to RAM, so it can not be used in high speed circuits.

memory

FLASH MEMORY

- This is a modified EEPROM. The difference is the erasure procedure.
- EEPROM can be erased at a register level, but flash memory must be erased either in its entirety or at the sector (block) level

Secondary Memory

- Secondary memory are storage devices. These devices have high data holding capacity.
- They store programs that are not frequently used by the processor.
- They are slow and have larger size

memory

Performance of memory

➤ Access time (t_a)

- Read access time: It is the average time required to read the unit of information from memory
- Write access time: It is the average time required to write the unit of information on memory
- Access rate $r_a = 1 / t_a$

➤ Cycle time (t_c)

- It is the average time that lapses between two successive read operation Cycle rate (r_c) = bandwidth = $(1/t_c)$

memory

➤ Access modes of memory

➤ Random access

- In random access mode, the t_a is independent of the location from which the data is accessed like MOS memory

➤ Sequential access

- In that mode, the t_a is dependent of the location from which the data is accessed like magnetic type

➤ Semi random access

- The semi random access combines these two For e.g. In magnetic disk, any track can be accessed at random But the access within the track must be in serial fashion

Memory hierarchy

- Capacity, cost and speed of different types of memory play a vital role while designing a memory system for computers.
- If the memory has larger capacity, more application will get space to run smoothly.
- It's better to have fastest memory as far as possible to achieve a greater performance. Moreover for the practical system, the cost should be reasonable.
- There is a tradeoff between these three characteristics cost, capacity and access time. One cannot achieve all these quantities in same memory module because
 - If capacity increases, access time increases (slower) and due to which cost per bit decreases.
 - If access time decreases (faster), capacity decreases and due to which cost per bit increases.
- The designer tries to increase capacity because cost per bit decreases and the more application program can be accommodated. But at the same time, access time increases and hence decreases the performance.

Memory hierarchy

- **So the best idea will be to use memory hierarchy.**
- Memory Hierarchy is to obtain the highest possible access speed while minimizing the total cost of the memory system.
- Not all accumulated information is needed by the CPU at the same time.
- Therefore, it is more economical to use low-cost storage devices to serve as a backup for storing the information that is not currently used by CPU
- The memory unit that directly communicate with CPU is called the main memory
- Devices that provide backup storage are called auxiliary memory
- The memory hierarchy system consists of all storage devices employed in a computer system from the slow by high-capacity auxiliary memory to a relatively faster main memory, to an even smaller and faster cache memory

Memory hierarchy

- The main memory occupies a central position by being able to communicate directly with the CPU and with auxiliary memory devices through an I/O processor
- A special very-high-speed memory called cache is used to increase the speed of processing by making current programs and data available to the CPU at a rapid rate
- CPU logic is usually faster than main memory access time, with the result that processing speed is limited primarily by the speed of main memory
- The cache is used for storing segments of programs currently being executed in the CPU and temporary data frequently needed in the present calculations
- The memory hierarchy system consists of all storage devices employed in a computer system from slow but high capacity auxiliary memory to a relatively faster cache memory accessible to high speed processing logic. The figure below illustrates memory hierarchy.

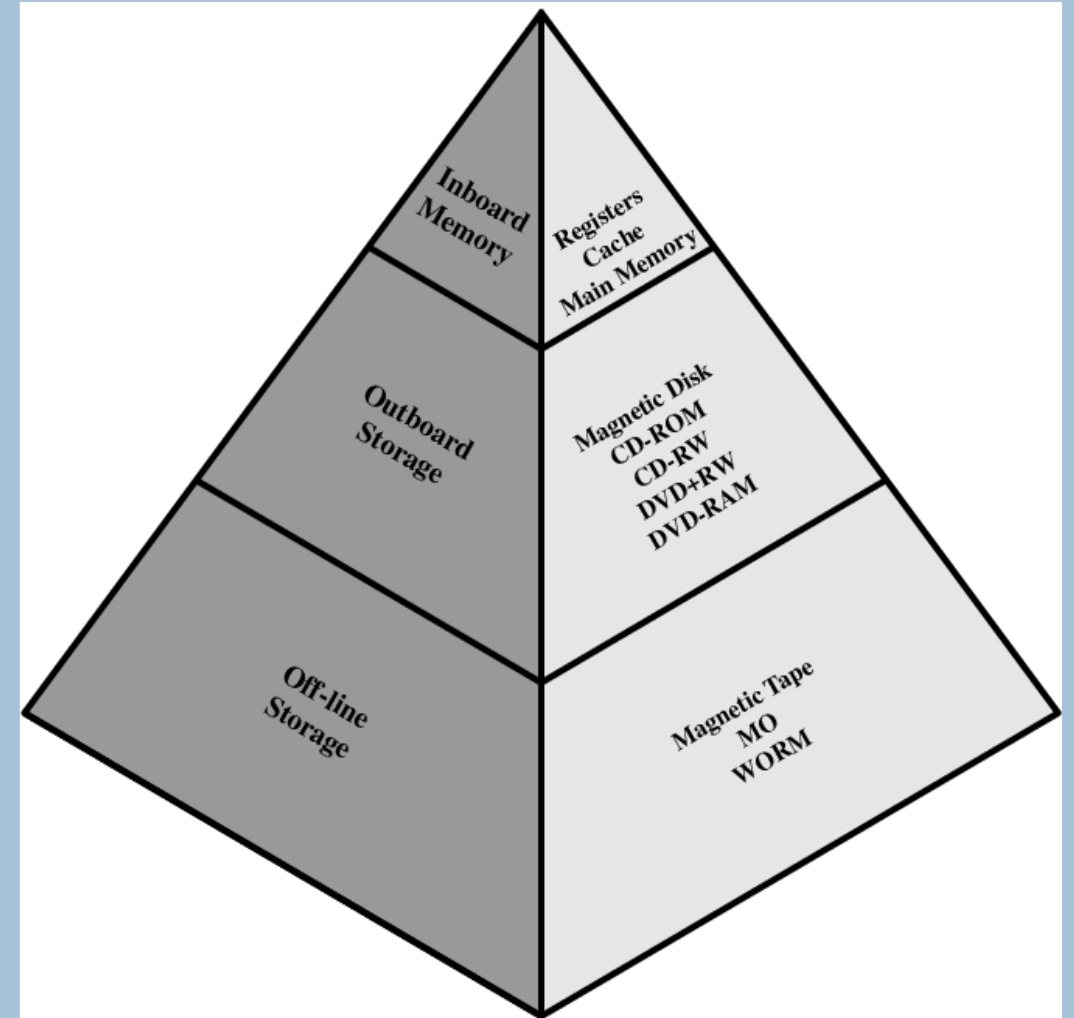
Memory hierarchy

As we go down in the hierarchy

- Cost per bit decreases
- Capacity of memory increases
- Access time increases
- Frequency of access of memory by processor also decreases.

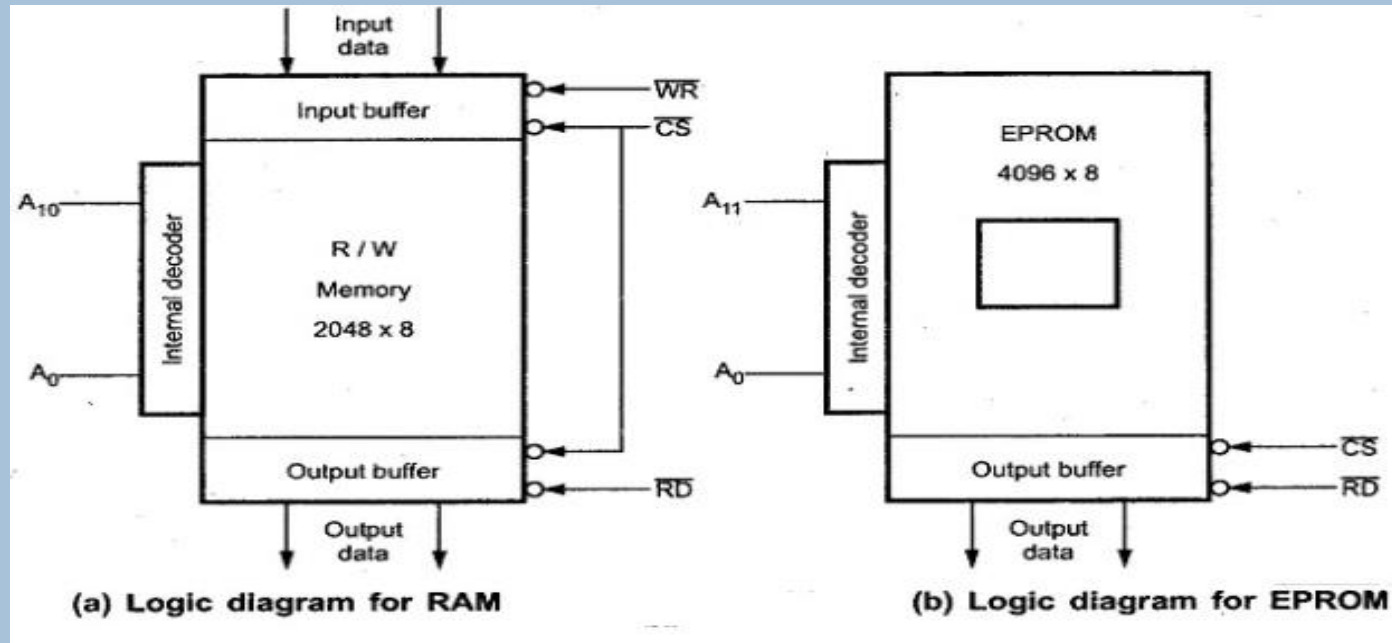
Hierarchy List

- Registers
- L1 Cache
- L2 Cache
- Main memory
- Disk cache
- Disk
- Optical
- Tape



Memory structure and its requirements

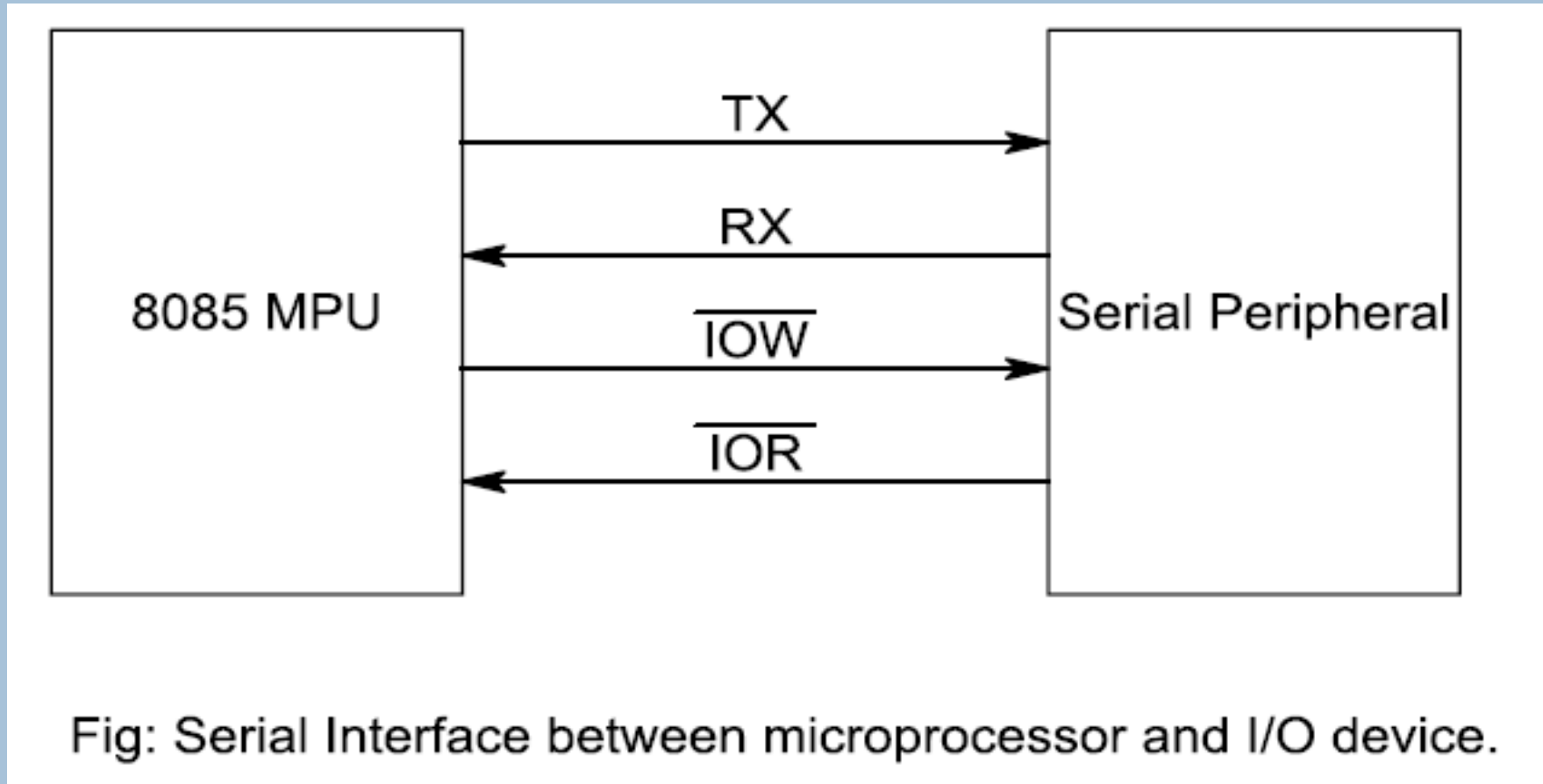
- Internally a memory consists of address decoder, input buffer, output buffer, registers with address lines, data lines, and (RD') (WR') , (CS') control lines.
- The number of address lines will be determined by the memory capability.
- The number of data lines will be determined by memory size.
- For $2^n \times m$ memory capability, the number of address line =n and the number of data lines =m.



Address decoding

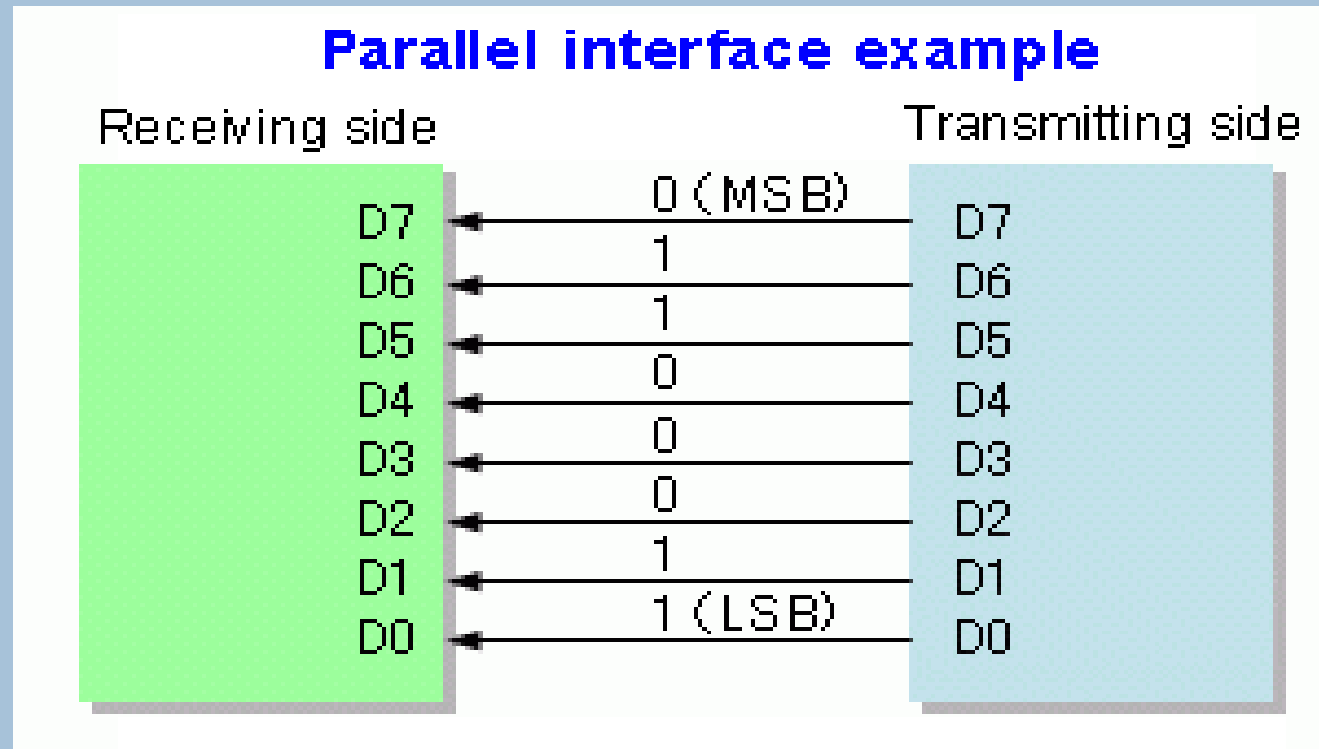
Memory mapped I/O	I/O mapped I/O
1. In this device address is 16 bit. Thus A_0 to A_{15} lines are used to generate device address.	1. In this I/O device address is 8 bit. Thus A_0 to A_7 or A_8 to A_{15} lines are used to generate device address.
2. $\overline{\text{MEMR}}$ and $\overline{\text{MEMW}}$ control signals are used to control read and write I/O operations.	2. $\overline{\text{IOR}}$ and $\overline{\text{IOW}}$ control signals are used to control read and write I/O operations.
3. Instructions available are LDA addr, STA addr, LDAX rp, STAX rp, MOV M,R, MOV R,M ADD M, CMP M etc.	3. Instructions available are IN and OUT.
4. Data transfer is between any register and I/O device.	4. Data transfer is between accumulator and I/O device.
5. Maximum number of I/O devices are 65536 (theoretically).	5. Maximum number of I/O devices are 256.
6. Execution speed using LDA addr, STA addr is 13 T-state and 7 T-states for MOV M, r and MOV r, M instructions.	6. Execution speed is 10 T-states.
7. Decoding 16 bit address may require more hardware.	7. Decoding 8 bit address will require less hardware.

Serial interface



Parallel interface

- The device which can handle data at higher speed cannot support with serial interface.
- N bits of data are handled simultaneously by the bus and the links to the device directly.
- Achieves faster communication but becomes expensive due to need of multiple wires.



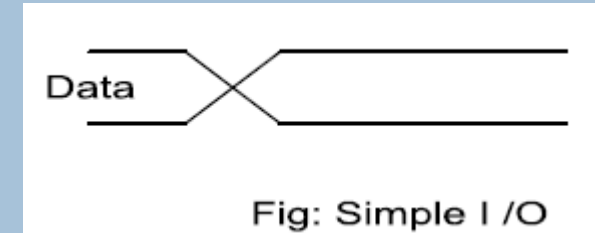
Synchronizing the computer with peripherals

- The information exchanged between a microprocessor and an I/O interface circuit consists of input or output data and control information.
- The status information enable the microprocessor monitor the device and when it is ready then send or receive data.
- Control information is the command by microprocessor to cause I/O device to take some action.
- If the device operates at different speeds, then microprocessor can be used to select a particular speed of operation of the device.
- The techniques used to transfer data between different speed devices and computer is called synchronizing. Different techniques under synchronizing are:

Synchronizing the computer with peripherals

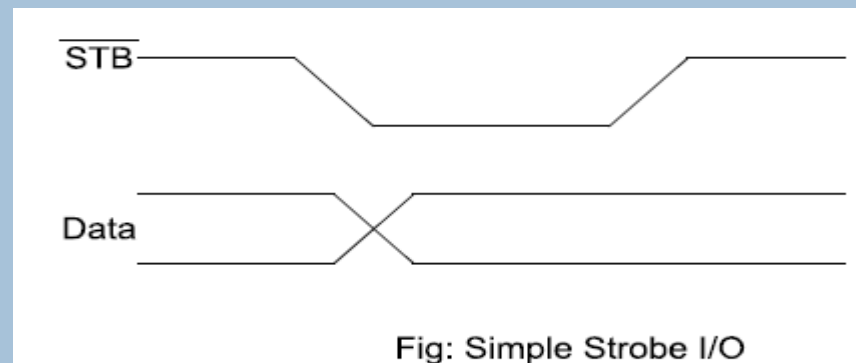
Simple I/O:

- For simple I/O, the buffer switch and latch switches i. e. LED are always connected to the input and output ports.
- The devices are always ready to send or receive data.
- Here cross line indicate the time for new valid data.



Wait Interface(Simple strobe I/O)

- In this technique, microprocessor need to wait until the device is ready for the operation.



Synchronizing the computer with peripherals

- Consider a simple keyboard consisting of 8 switches connected to a microprocessor through a parallel interface circuit (Tri-state buffer).
- The switch is of dip switches.
- In order to use this keyboard as an input device the microprocessor should be able to detect that a key has been activated.
- This can be done by observing that all the bits are in required order.
- The processor should repeatedly read right order of bits i.e. at least 1 bit of 8
- Consider the tri-state A/D converter

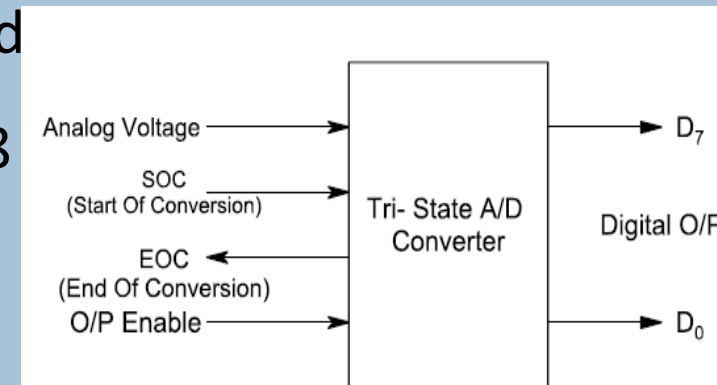


Fig: Tri-State A/D Converter

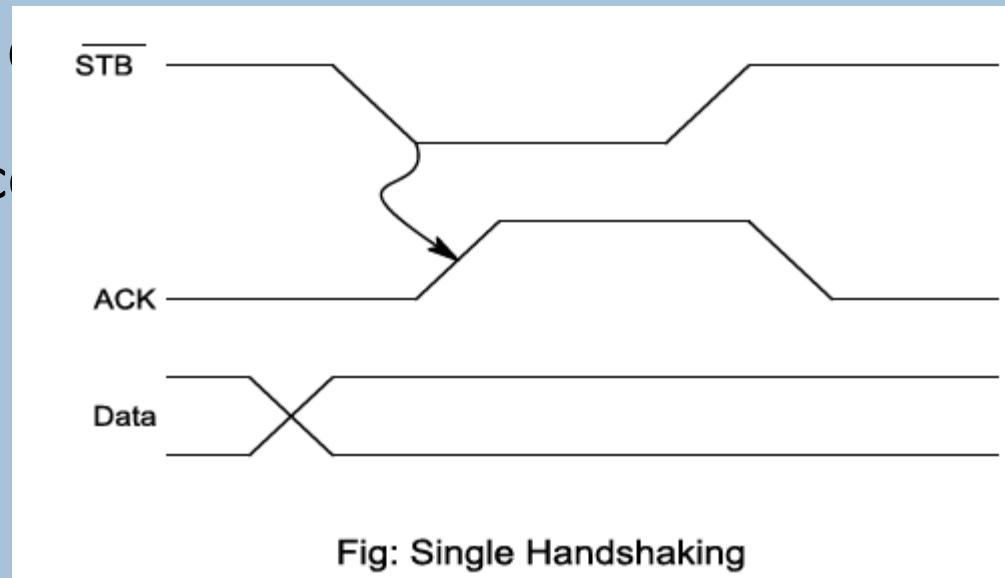
Synchronizing the computer with peripherals

- Used to convert analog to digital data which can be read by I/O unit of MP
- When SOC appears 1, I/O unit should be ready for reading binary data/digital data.
- When EOC's status is 1, then I/O unit should stop to read data.
- Strobe signal indicates the time at which data is being activated to transmit.

Synchronizing the computer with peripherals

Single Handshaking:

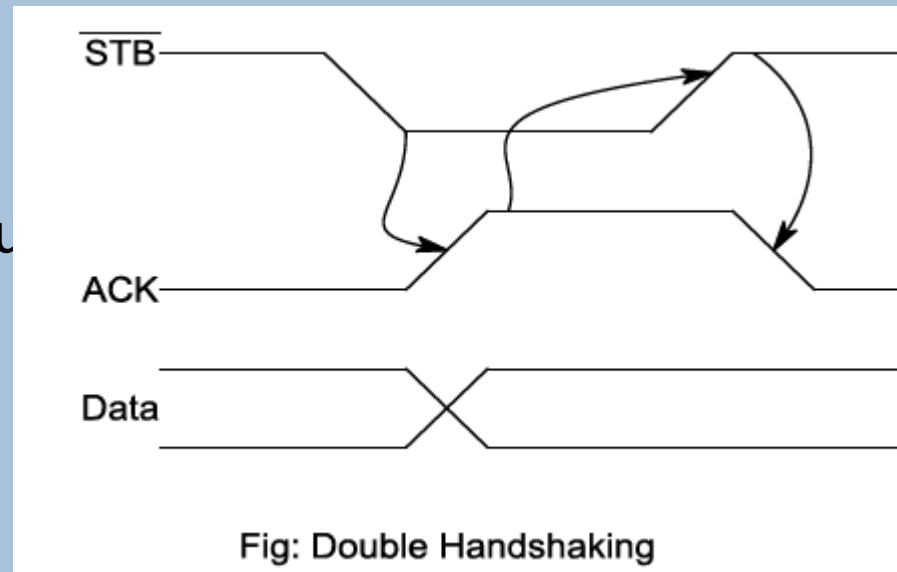
- The peripheral outputs some data and send $\overline{STB}$ signal to microprocessor. “here is the data for you.”
- Microprocessor detects asserted $\overline{STB}$ signal, reads the data and sends an acknowledge signal (ACK) to indicate data has been read and peripheral can send next data.
- Microprocessor sends next data to peripheral. Peripheral is ready.



Synchronizing the computer with peripherals

Double Handshaking:

- The peripheral asserts its STB' line low to ask MP "Are you ready?"
- The MP raises its ACK line high to say " I am ready".
- Peripheral then sends data and raises its STB' line low to say "Here is some valid data for you."
- MP then reads the data from the peripheral, and I await your request.



ave the data, thank

Programmable peripheral interface (ppi) – 8255A

- The Intel 8255A is a general purpose programmable I/O device designed for use with Intel microprocessors.
- It has 24 I/O pins that can be grouped primarily in two 8-bit parallel ports: A and B, with the remaining bits as port C.
- The 8-bits of port C can be used as individual bits or be grouped in two 4-bits ports: C_{upper} (C_u) and C_{lower} (C_l).
- The functions of these ports are defined by writing a control word in the control register.

Programmable peripheral interface (ppi) – 8255A

Block diagram:

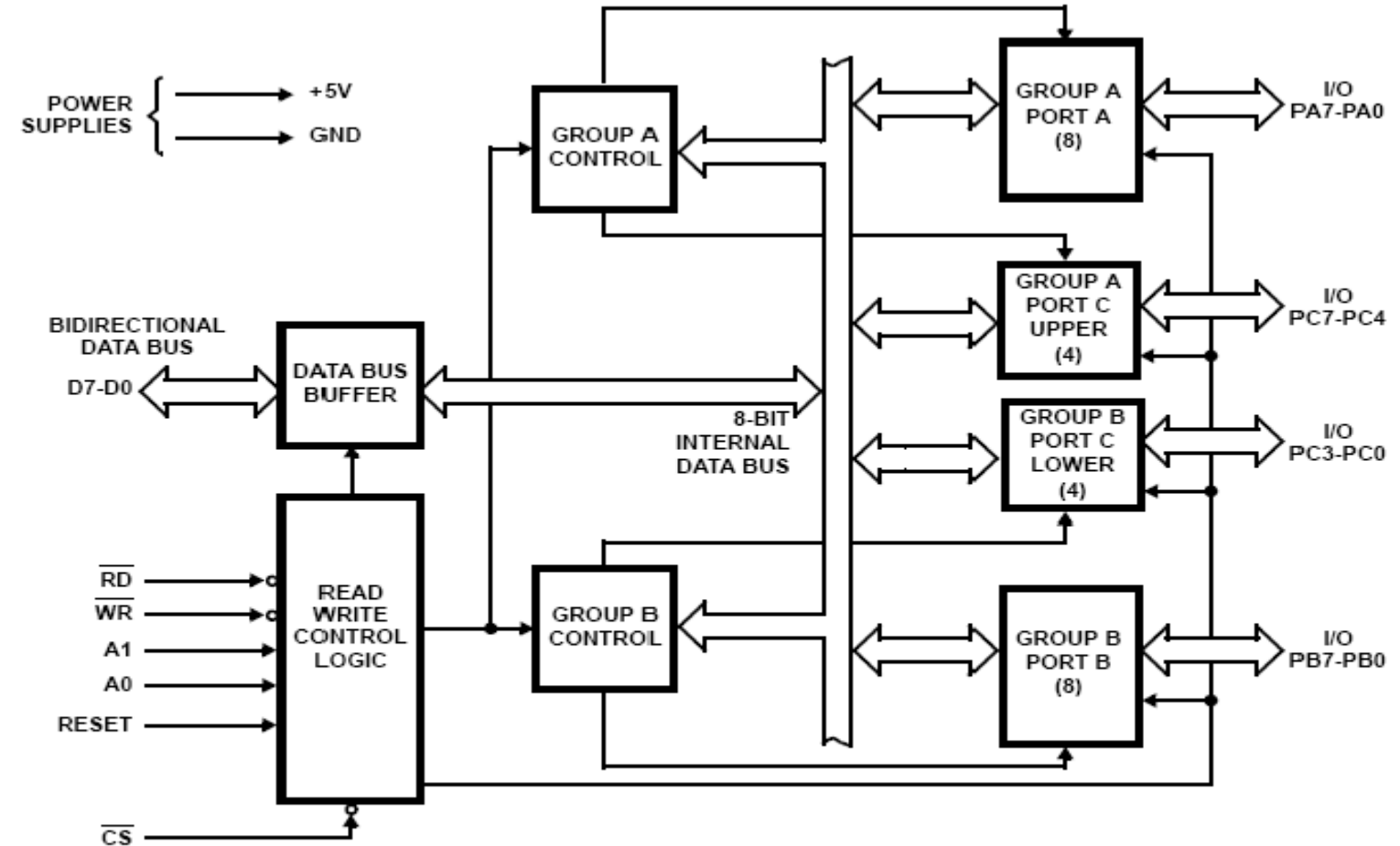


Fig2: Internal Block Diagram of 8255

Programmable peripheral interface (ppi) – 8255A

Data Bus Buffer

- The 3-state bidirectional 8-bit buffer is used to interface the 8255A to the system data bus.
- Data is transmitted or received by the buffer upon execution of input or output instructions by the CPU.
- Control words and status information are also transferred through the data bus buffer.

Read/Write Control Logic

- The function of the block is to manage all of the internal and external transfers of both data and control or status words.
- It accepts inputs from the CPU address and control buses and in turn, issues commands to both of the control groups.
- Chip Select (CS'): A “low” on this pin enables the communications between the 8255A and the CPU.

Programmable peripheral interface (ppi) – 8255A

- **Read (RD')**: A “low” on this input enables the 8255A to send the data or status information to the CPU on the data bus. In essence, it allows the CPU to read from the 8255A.
- **Write (WR')**: A “low” on this input pin enables the CPU to write data or control words into the 8255A.
- **Reset (RESET)**: A “high” to this pin clears the control register and sets all ports (A, B and C) in the input mode.
- **A0 and A1**: These input signals controls the selection of one of the three ports or the control word register. They are connected to the least significant bits of the address bus.
- The **CS'** signal is the master chip select, and A0 and A1 specify one of the I/O ports or the control register as given below.

Programmable peripheral interface (ppi) – 8255A

CS'	A ₁	A ₀	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	X	X	8255A is not selected

➤ Group A and Group B controls

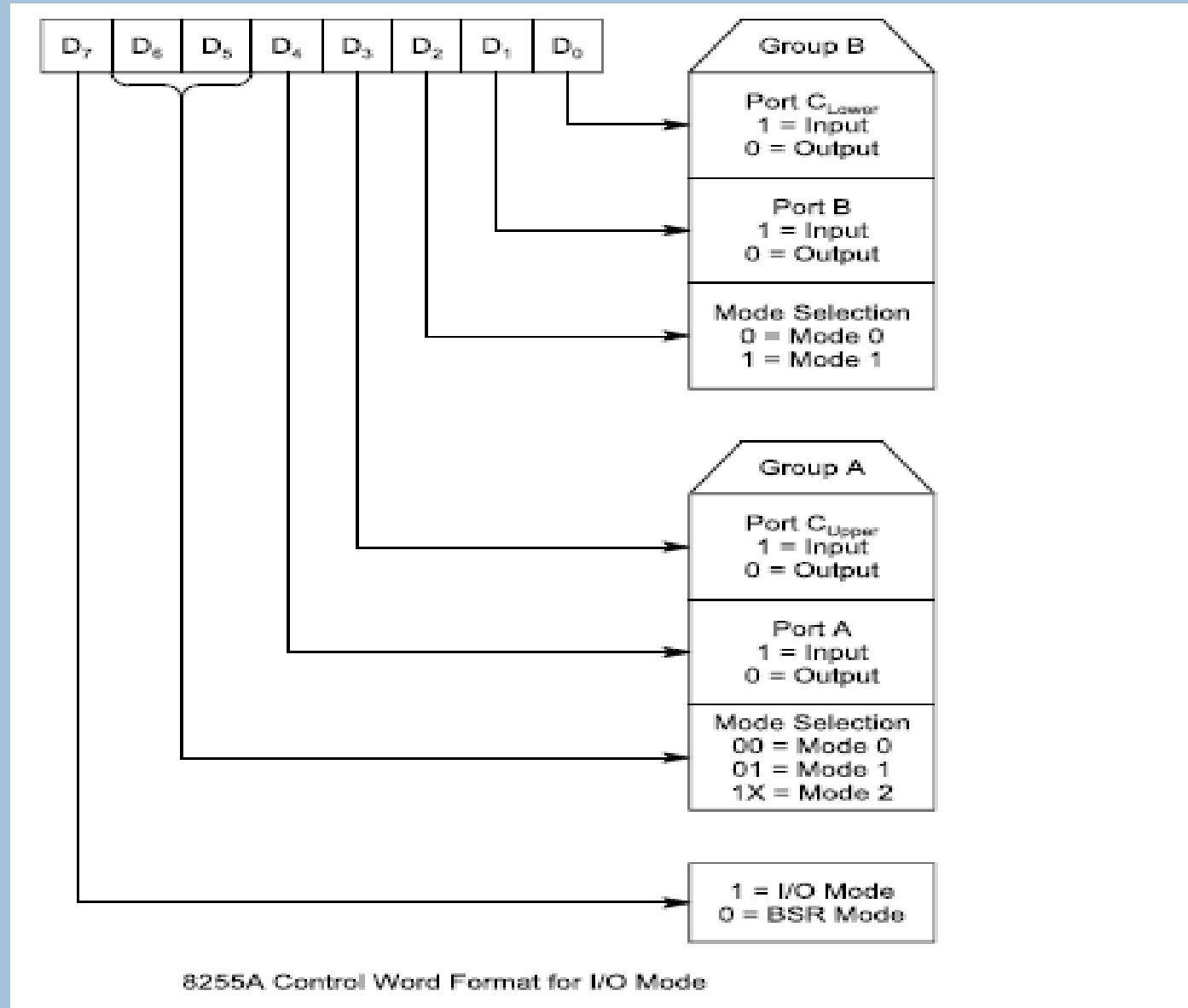
- Each of the control blocks (Group A and Group B) accepts “commands” from the Read/Write control logic, receives control word from the internal data bus and issues the proper commands to its associated ports.
- Control Group A – Port A and Port C_{Upper} (C7 – C4)
- Control Group B – Port B and Port C_{Lower} (C3 – C0)

Programmable peripheral interface (ppi) – 8255A

Control Word

- When A0 and A1 pins have value 1, the mapped address addresses the control register which is the 8-bit register to write the specific content according to the port conditions although it cannot be read. The content of this register is called control word which specifies an I/O function for each port.
- To communicate with peripherals through 8255, following steps are necessary.
- Determine the Port addresses of Ports A, B and C and of the control register, according to Chip Select logic and address lines A1 and A0.
- Write a control word in control register.
- Write I/O instructions to communicate with peripherals through Ports A, B and C.

Programmable peripheral interface (ppi) – 8255A



Programmable peripheral interface (ppi) – 8255A

Operating modes:

- **Bit Set/Reset mode:** The BSR mode is used to set or reset the bits in port C.
- **I/O mode:** The I/O mode is further divided into three modes: mode 0, mode 1 and mode 2.
- In **mode 0**, all ports function as simple I/O ports.
- **Mode 1** is a handshake mode whereby ports A and/or B use bits from port C as handshake signals. In the handshake mode, two types of I/O data transfer can be implemented: status check and interrupt.
- In **mode 2**, port A can be set up for bidirectional data transfer using handshake signals from port C and port B can be set up either in mode 0 or mode 1.

Programmable peripheral interface (ppi) – 8255A

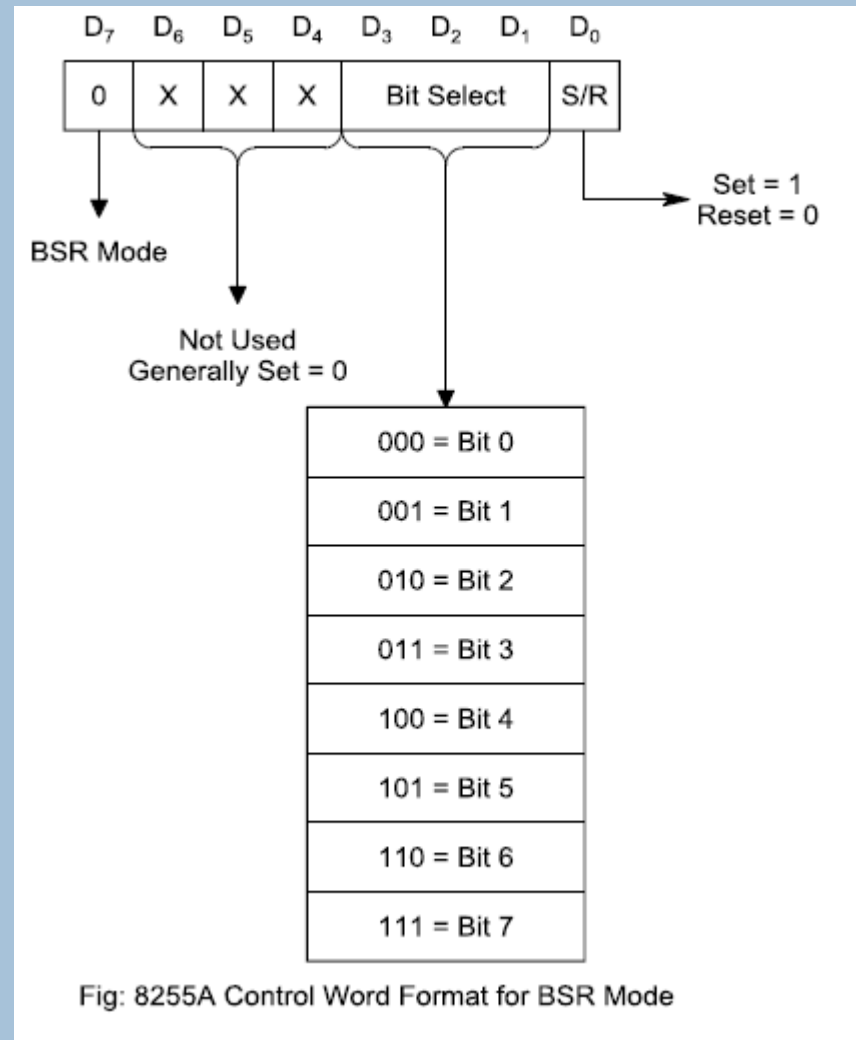
BSR Mode (Bit Set/Reset)

- BSR mode is concerned only with eight bits of port C, which can be set or reset by writing an appropriate control word in the control register.
- A control word with bit D7=0 is recognized as a control word and it does not alter any previously transmitted control word with bit D7=1; thus the I/O operations of ports A and B are not affected by a BSR control word.
- In the BSR mode individual bits of port C can be used for applications such as On/Off switch

BSR Control Word:

- This control word, when written in control register, sets or resets one bit at a time, as specified in figure

Programmable peripheral interface (ppi) – 8255A



Programmable peripheral interface (ppi) – 8255A

Mode 0 (Basic Input/output)

- This functional configuration provides simple input and output operation for each of the three ports.
No ‘handshaking’ is required; data is simply written to or read from a specified port..
- Mode 0 basic functional definitions:
 - Two 8-bit ports and two 4-bit ports
 - Any port can be input or output
 - Outputs are latched
 - Inputs are not latched
 - 16 different input/output configurations are possible in this mode.

Programmable peripheral interface (ppi) – 8255A

Mode 1 (Strobe Input/output)

- The functional configuration provides a means for transferring I/O data to or from a specified port in conjunction with strobes or handshaking signals.
- In mode 1, port A and port B use the lines of port C to generate or accept these handshaking signals.
- Mode 1 basic functional definitions:
 - Two groups (Group A and Group B)
 - Each group contains one 8-bit data port and one 4-bit control/data port
 - The 8-bit data port can be either input or output. Both inputs and outputs are latched.
 - The 4-bit port is used for control and status of the 8-bit data port.

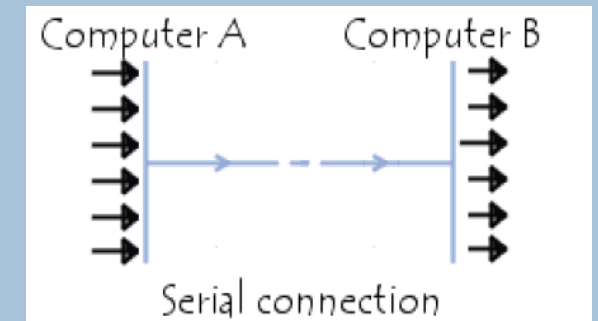
Programmable peripheral interface (ppi) – 8255A

Mode 2 (Strobe Bidirectional Bus I/O)

- The functional configuration provides a means for communicating with a peripheral device or a structure on a single 8-bit bus for both transmitting and receiving data (bidirectional bus I/O).
- “Handshaking Signals” are provided to maintain proper bus flow discipline in a similar manner to Mode 1.
- Interrupt generation and enable/disable functions are also available.
- Mode 2 basic functional definitions:
 - Used in Group A only
 - One 8-bit bidirectional bus port (Port A) and a 5-bit control port (Port C)
 - Both inputs and outputs are latched
 - The 5-bit control port (Port C) is used for control and status for the 8-bit, bidirectional bus port (Port A)

Serial data transmission

- Data are sent one bit at a time over the transmission channel.
- However, since most processors process data in parallel, the transmitter needs to transform incoming parallel data into serial data and the receiver needs to do the opposite.
- Cost of communication hardware is considerable reduced since only a single wire or channel is required for transmission.
- Slow as compared to parallel transmission.
- Serial data can be sent synchronously or asynchronously.



Serial data transmission - advantages

- Data transmission over longer distance because voltage loss is not much a problem in serial communication.
- Serial; 1 → -3V to -25V
- 0 → +3V to +25V
- Parallel; 1 → +5V
- 0 → 0V
- Requires less number of wires than parallel and so cheaper to transmit data.
- Crosstalk is less of an issue because there are fewer conductors' compared to that of parallel cables.
- Many IC and peripherals have serial interface
- Clock skew between different cables is not an issue
- Serials can be clocked at higher data rate
- Serial cable can be longer than parallel
- Cheaper to implement
- *But in serial mode of transfer, only one bit of a word is transferred at a time so that data transfer rate is very slow; it is the one of the demerit over parallel data transfer*

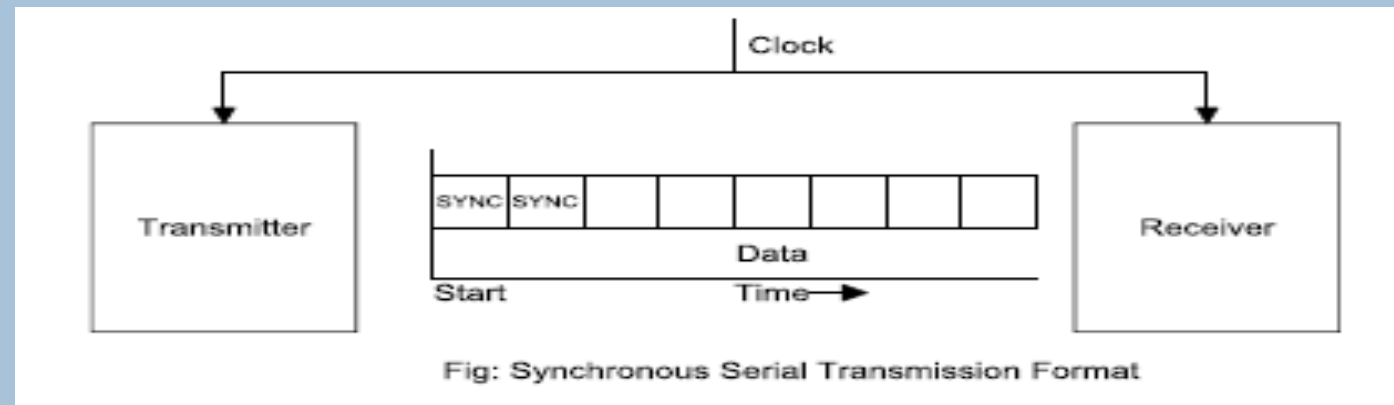
Serial data transmission

Serial Synchronous Data Transmission

- Data is transmitted or received based on a clock pulse (i.e. one bit at each clock pulse)
- In order to interpret the data correctly, the receiving device must know the start and end of each data unit.
- The transmitter must know the number of data units to be transferred and the receiver must be synchronized with the data boundaries.
- Therefore, there must be synchronization between the transmitter and receiver.
- Usually one or more SYNC characters are used to indicate the start of each synchronous data stream or frame of data.
- Transmitter sends a large block of data characters one after the other with no time between characters.

Serial data transmission

- Transmitting device sends data continuously to the receiving device.
- If the data is not ready to be transmitted, the line is held in marking condition.
- To indicate the start of transmission, the transmitter sends out one or more SYNC characters or a unique bit pattern called a flag, depending on the system being used.
- The receiving device waits for data, when it finds the SYNC characters or the flag then starts interpreting the data which shifts the data following the SYNC characters and converts them to parallel form so they can be read in by a computer.



Serial data transmission

➤ **Advantages and Disadvantages of Synchronous Communication**

➤ Main advantage of Synchronous data communication is the high speed.

The synchronous communications require high-speed peripherals/devices and a good-quality, high bandwidth communication channel.

➤ The disadvantage includes the possible in accuracy. Because when a receiver goes out of Synchronization, loosing tracks of where individual characters begin and end. Correction of errors takes additional time.

Serial data transmission

Serial Asynchronous Data Transmission

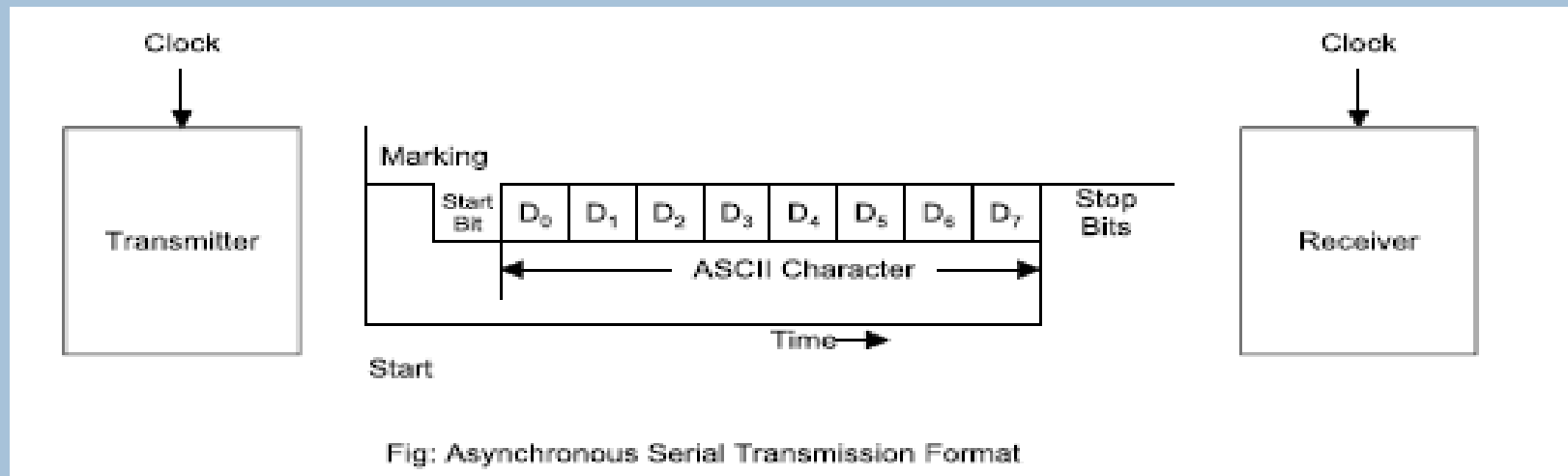
- The receiving device does not need to be synchronized with the transmitting device.
- The transmitting device can send one or more data units when it is ready to send data.
- Each data unit must be formatted i.e. must contain start and stop bits for indicating beginning and the end of data unit. It also includes one parity bit to identify odd or even parity of data.
- To send ASCII character, the framing of data should contain:
 - 1 start bit: Beginning of data
 - 8 bit character: Actual data transferred
 - 1 or 2 stop bits: End of data
- When no data is being sent, the signal line is in a constant high or marking state.

Serial data transmission

- The beginning of the data character is indicated by the line going low for 1 bit time and this bit is called a start bit.
- The data bits are then sent out on the line one after the other where the least significant bit is sent out first.
- Parity bit should contain to check for errors in received data.
- After the data bit and a parity bit, the signal line is returned high for at least 1 bit time to identify the end of the character, this always high bit is referred to as a stop bit. Some older systems use 2 stop bits.
- Asynchronous communication is used when slow speed peripherals communicate with the computer.
- The main disadvantage of asynchronous communication is slow speed transmission.

Serial data transmission

- Asynchronous communication however, does not require the complex and costly hardware equipment's as is required for synchronous transmission.



Serial data transmission

➤ Synchronous versus Asynchronous serial data transmission

S.N.	Parameter	Asynchronous	Synchronous
1.	Fundamental	Transmission does not based on clock signal	Transmission based on clock signal
2.	Data Format	One character at a time	Group of characters i.e. a block of characters
3.	Speed	Low (< 20 kbps)	High (> 20 kbps)
4.	Framing Information	Start and stop bits are sent with each character.	SYNC characters are sent with each character.
5.	Implementation	Hardware / Software	Hardware

Bit rate and baud rate

➤ Bit Rate:

- Measure of no of data bits transmitted per sec.
- E.g. 2400 bits per sec means 2400 zeros or ones can be transmitted in one sec.

➤ Baud Rate:

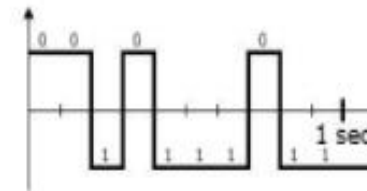
- No of times a signal in a communication channel changes state.
- Change state means change from 0 to 1 or from 1 to 0 up to 2400 times per sec.

- E.g. 2400 baud rate means “the channel can change states up to 2400 time per sec”

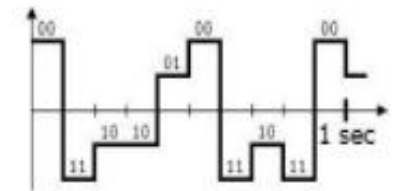
Baud → How many times a signal changes per second

Bit rate → How many bits can be sent per time unit (usually per second)

Bit rate is controlled by baud and number of signal levels



Baud = 10
Bit rate = 10 bps



Baud = 10
Bit rate = 20 bps

Standards in serial i/o

- The serial I/O technique is commonly used to interface different peripheral terminals such as printers, modems with microcomputers which are designed and manufactured by various manufacturers.
- Therefore, a common understanding must exist, among various manufacturing and user groups that can ensure compatibility among different equipment.
- The standard is defined as the understanding which is accepted in industry and by users.
- A standard is normally defined by a professional organizations such as IEEE (Institute of Electrical and Electronics Engineers), EIA (Electronic Industries Association) as a de jure standard. However, a widespread practice can become a de facto standard.
- In serial I/O, data can be transmitted as either current or voltage.

Rs-232c

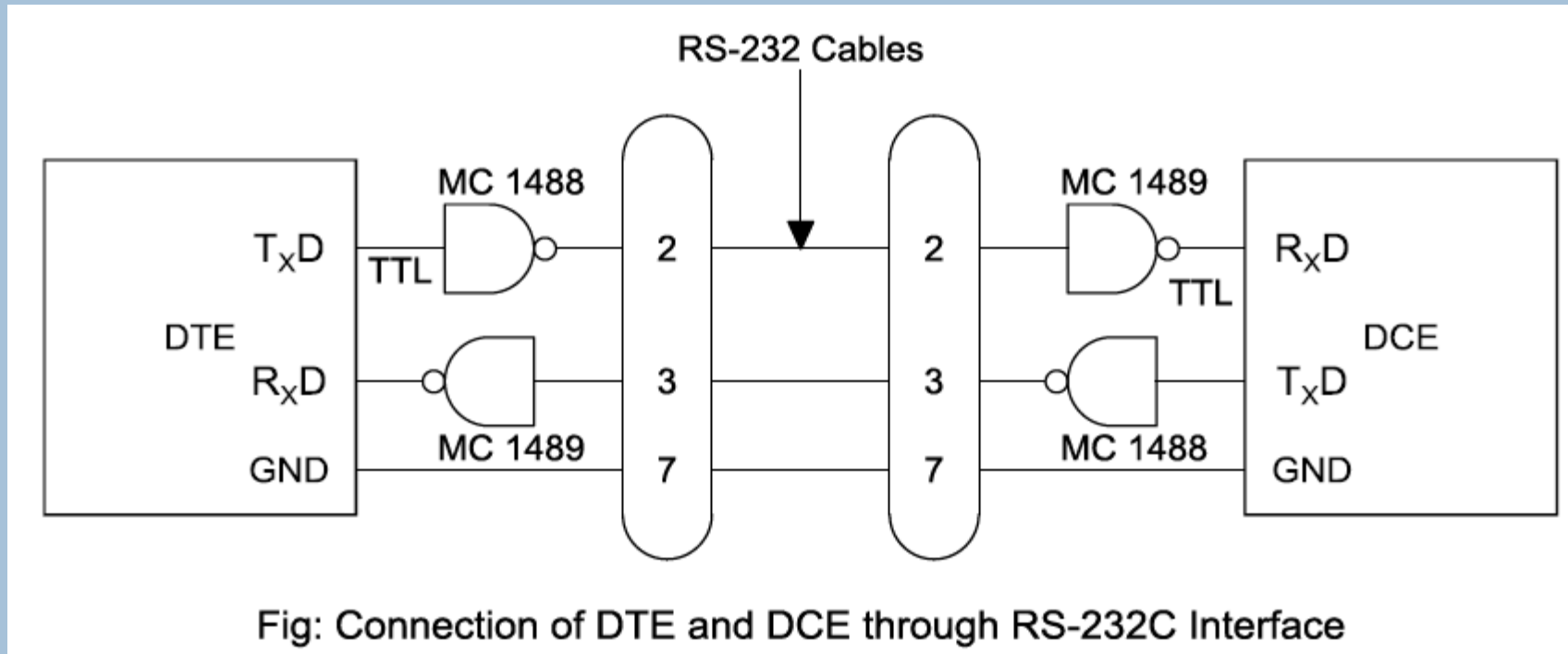
- RS-232C is an interface developed to standardize the interface between data terminal equipment (DTE) and data communication equipment (DCE) employing serial binary data exchange.
- Modem and other devices used to send serial data are called data communication equipment (DCE).
- The computers or terminals that are sending or receiving the data are called data terminal equipment (DTE).
- RS- 232C is the interface standard developed by electronic industries Association (EIA) in response to the need for the signal and handshake standards between the DTE and DCE.

Rs-232c

RS-232C has following standardize features.

- It uses 25 pins (DB – 25P) or 9 Pins (DE – 9P) standard, where 9 pins standard does not use all signals i.e. data, control, timing and ground.
- It describes the voltage levels, impedance levels, rise and fall times, maximum bit rate and maximum capacitance for all signal lines.
- It specifies that DTE connector should be male and DCE connector should be female.
- It can send 20kBd for a distance of 50 ft.
- The voltage level for RS-232 are:
 - A logic high or 1 or mark, -3V to -15V
 - A logic low or 0 or space, +3v to +15v
- Normally $\pm 12V$ voltage levels are used

Rs-232c



- Mc1488 line driver converts logic 1 to -9V Logic 0 to +9v
- Mc1489 line receiver converts RS – 232 to TTL
- Signal levels of RS-232 are not compatible with that of the DTE and DCE which are TTL signals for that line driver such as M 1488 and line receiver MC1489 are used.

RS- 232 signals used in handshaking:

Signal Flow	DE-9P	DB-25P	Signal	Description
-	1	-	Protective Ground	-
DTE to DCE	3	2	TxD	Transmitted Data
DCE to DTE	2	3	RxD	Received Data
DTE to DCE	7	4	$\overline{RTS}$	Request To Send
DCE to DTE	8	5	$\overline{CTS}$	Clear To Send
DCE to DTE	6	6	$\overline{DSR}$	Data Set Ready
Common Ref	5	7	GND	Signal Ground
DCE to DTE	1	8	$\overline{DCD}$	Data Carrier Detect
DTE to DCE	4	20	$\overline{DTR}$	Data Terminal Ready
DCE to DTE	9	22	RI	Ring Indicator
DCE to DTE	-	23	DSRD	Data Signal Rate Detector

Rs-232c

➤ **Data Terminal Ready (DTR):**

- After the terminal power is turned on and terminal runs any self checks, it asserts data terminal ready (DTR') signal to tell the modem that it is ready.

➤ **Data Set Ready (DSR):**

- When the MODEM is powered up and ready to transmit or receive data, it will assert data set ready (DSR') to the terminal. Under manual control or terminal control, modem then dials up the computer. If the computer is available, it will send back a specified tone.

➤ **Request to send (RTS):**

- When a terminal has a character ready to send, it will assert a request-to-send (RTS') signal to the modem.

➤ **Data Carrier Detect (DCD):**

- The modem will then assert its data-carrier-detect (DCD') signal to the terminal to indicate that it has established connection with the computer.

Rs-232c

➤ **Clear to send (CTS):**

- When the modem is fully ready to receive data, it asserts the clear-to-send (CTS') signal back to the terminal.

➤ **Ring indicator (RI):**

- It indicates that a ring has occurred at modem. Deactivating DTR or DSR breaks the connection but RI works independently of DTR i.e. a modem may activate RI signal even if DTR is not active.

➤ **Transmitted Data (TxD):**

- The terminal then sends serial data characters to the modem.

➤ **Received Data (RxD):**

- Modem will receive data from terminal through this line.

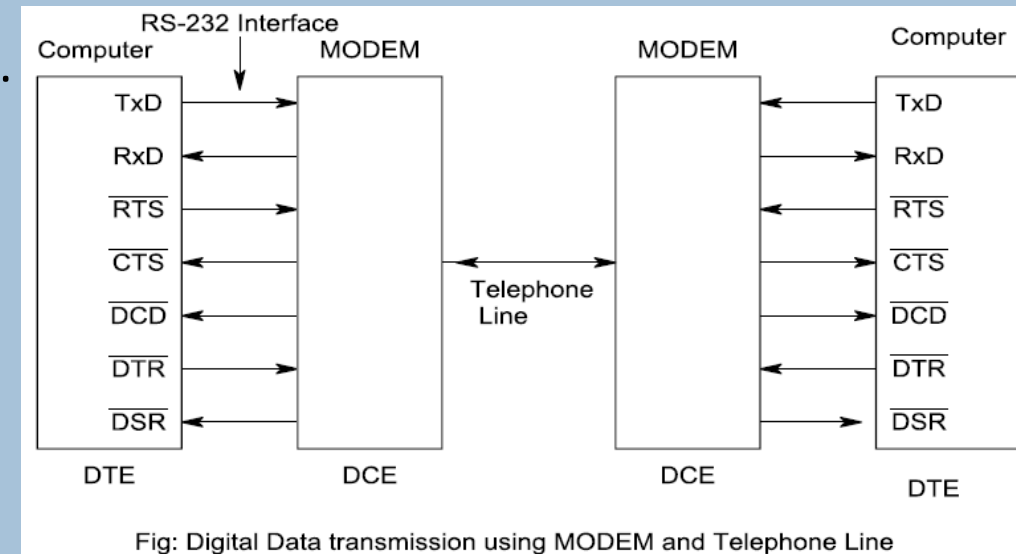
➤ **Data Signal Rate Detect (DSRD):**

- It is used for switching different baud rate.

Rs-232c

Digital Data Transmission Using Modem and standard Phone Lines

- Standard telephone system can be used for sending serial data over long distances.
- However, telephone lines are designed to handle voice, bandwidth of telephone lines ranges from 300 HZ
- to 3400 HZ.
- Digital signal requires a bandwidth of several megahertz. Therefore, data bits should be converted into audio tones, this is accomplished through modems.



Rs-232c

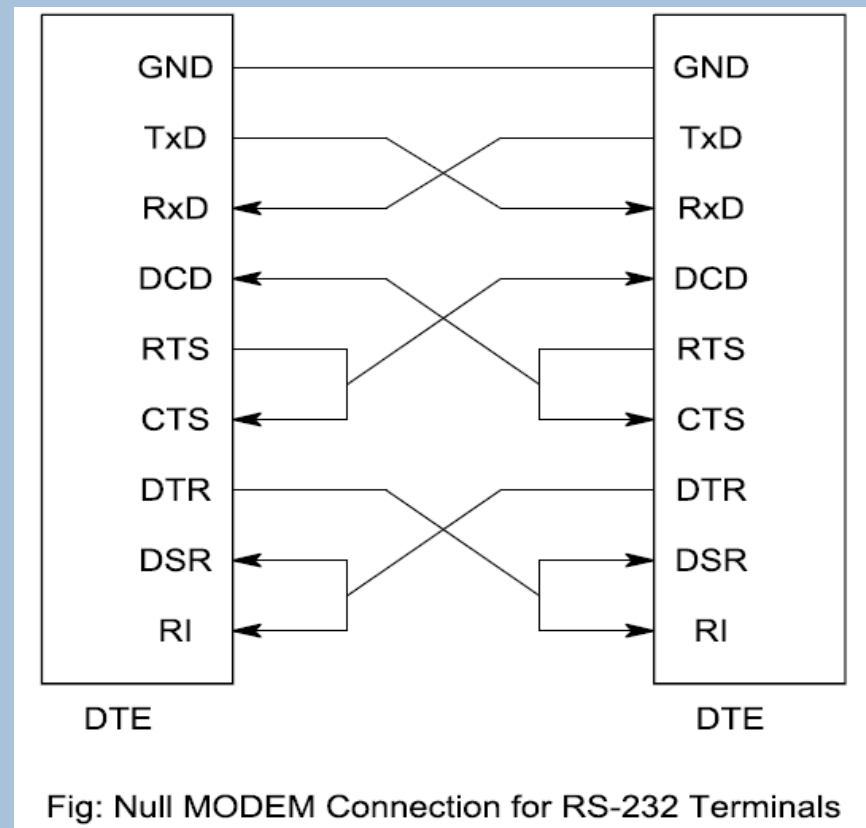
- DTE asserts DTR' to tell the modem it is ready.
- Then DCE asserts DSR' signal to the terminal and dials up.
- DTE asserts RTS' signal to the modem.
- Modem then asserts DCD' signal to indicate that it has established connection with the computer.
- DCE asserts CTS' signals, then DTE sends serial data.
- When sending completed, DTE asserts RTS' high, this causes modem to un assert its CTS' signal and stop transmitting similar handshake taken between DCE and DTE other side.
- To communicate from serial port of a computer to serial port of another computer without modem, null-modem is used.

Rs-232c – null modem connection

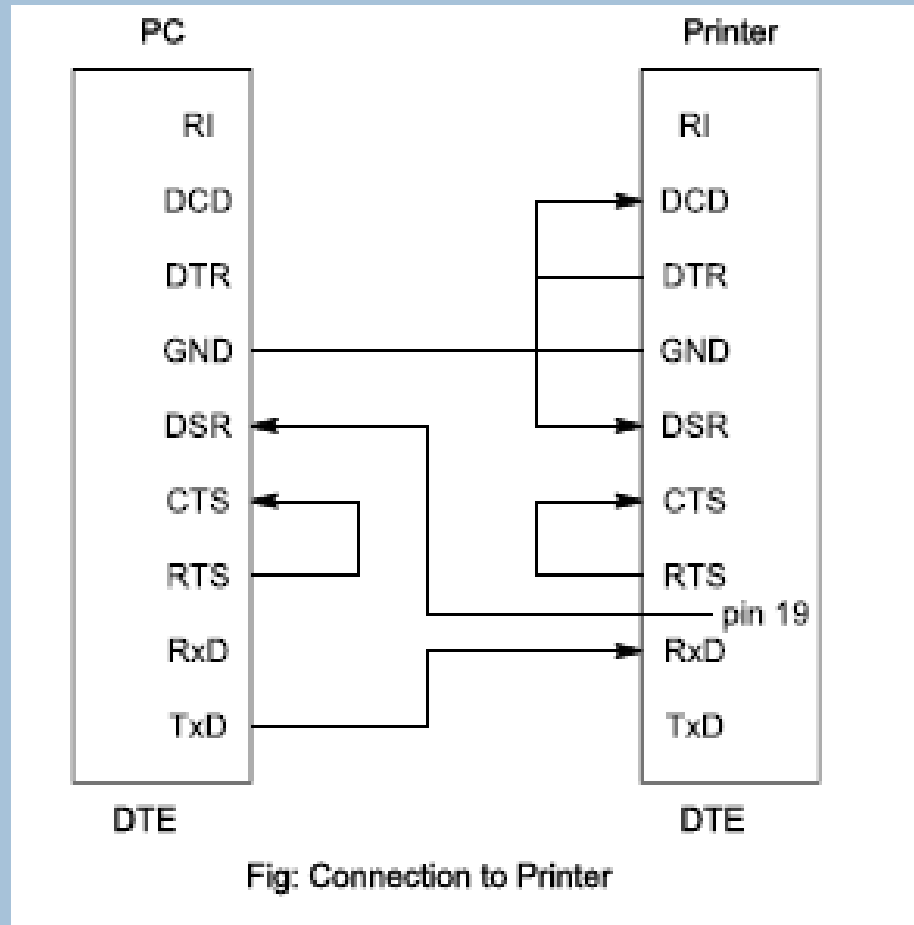
- A zero modem serves for data exchange between DTEs.
- Since both the computers are configured as DTEs, directly connecting them by means of the conventional serial interface cable is impossible; not even the plug fits into the jack of the second terminal.
- Also the TxD meets TxD and RxD meets RxD, DTR meets DTR and DSR meets DSR etc.
- This means that outputs are connected to outputs and inputs are connected to inputs. With this convention, no data transfer is possible.
- For the transmission of data, it is required to twist the TxD and RxD lines.
- In this way, the transmitted data of one terminal (PC) becomes received data of other and vice versa.
- As shown in figure, activation of RTS to begin a data transfer gives rise to an activation of CTS on same DTE and to an activation of DCD on other DTE.

Rs-232c – null modem connection

- Further, an activation of DTR leads to rise of DSR and RI on other DTE. Hence for every DTE, it is simulated that a DCE is on the end of line, although a connection between two DTEs is actually present.
- Zero modem can be operated with standard BIOS and DOS functions.



Rs-232c – connection to printer



- PC may send data faster than the printer can acknowledge it.
- Therefore, pin 19 (Buffer Full) of printer is connected to DSR of PC side.

Rs-232c – connection to printer

- An overflow of data deactivates the DSR signal and communication halts
- On PC RTS and CTS are connected to each other so that a transmission request from PC immediately enables the transmission.
- Printer as DTE refers to print anything as long as no active signal is present at inputs. Of CTS, DSR and DCD.
- This problem is resolved by connecting RTS with CTS and DTR with DCD and DSR.

Rs-423a

- A major problem with RS-232C is that it can only transmit data reliably for about 50 ft at its maximum rate of 20Kbd.
- If longer lines are used the transmission rate has to be drastically reduced due to open signal lines with a common signal ground.
- Another EIA standard which is improvement over RS-232C is RS-423A.
- The standardize features of RS-423 are:
 - This standard specifies a low impedance single ended signal which can be sent over 50 ohm coaxial cable and partially terminated at the receiving end to prevent reflection.
 - Voltage levels
 - Logic High 4V - 6V negative
 - Logic Low 4V - 6V positive
 - It allows a maximum data rate of 100 Kbd over 40 ft line or a maximum baud rate of 1 Kbd over 4000 ft line.

Rs-422a

It is a newer standard for serial data transfer. It specifies that each signal will be sent differentially over two adjacent wires in a ribbon cable or a twisted pair of wires uses differential amplifier to reject noise.

- The term differential in this standard means that the signal voltage is developed between two signal lines rather than between signal line and ground as in RS-232C and RS-423A.
- Any electrical noise induced in one signal line will be induced equally in the other signal line.
- A differential line receiver MC3486 responds only to the voltage difference between its two inputs so any noise voltage that is induced equally on two inputs will not have any effect on the output of the differential receiver.
- RS-422A has following standardized features:
 - Logic high is transmitted by making 'b' line more positive than 'a' line.
 - Logic low is transmitted by making 'a' line more positive than 'b' line.
 - The voltage difference between the two lines must be greater than 0.4V but less than 12V.

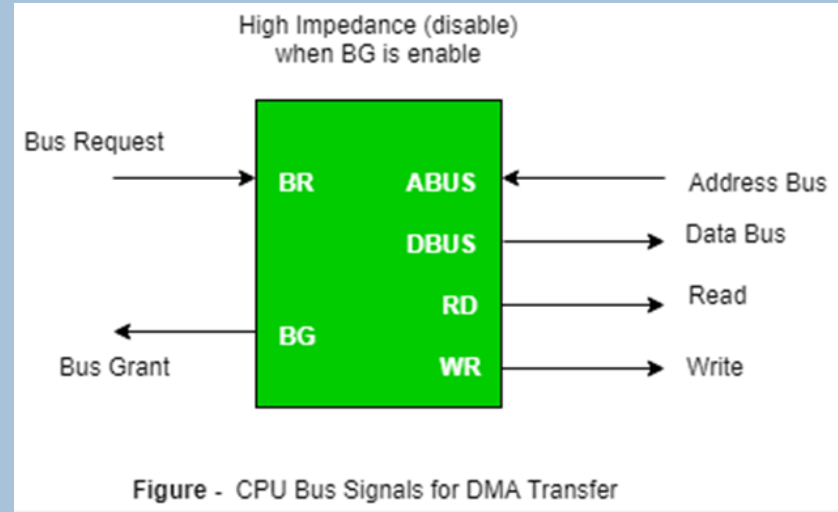
comparison

Comparison of Serial I/O Standards

S.N.	Specifications	RS-232C	RS-423A	RS-422A
1.	Speed	20 Kbaud	100 Kbaud at 40 ft 1 kbaud at 4000 ft	10 Mbaud at 40 ft 100 kbaud at 4000 ft
2.	Distance	50 ft	4000 ft	4000 ft
3.	Logic 0	+3 V to +25 V	+4 V to +6 V	B line > A line
4.	Logic 1	-3 V to -25 V	-4 V to -6 V	A line > B line
5.	Receiver Input Voltage	±15V	±12V	±7V
6.	Mode of Operation	Single ended input and output	Differential input and single ended output	Differential input and output
7.	Noise Immunity	2.0 V	3.4 V	1.8 V
8.	Input Impedance	3-7 KOhm and 2500 pf	>4 KOhm	>4 KOhm
9.	Short circuit current	500 mA	150 mA	150 mA

DMA (DIRECT MEMORY ACCESS)

- The data transfer between a fast storage media such as magnetic disk and memory unit is limited by the speed of the CPU
- Thus we can allow the peripherals directly communicate with each other using the memory buses, removing the intervention of the CPU. This type of data transfer technique is known as DMA or direct memory access
- During DMA the CPU is idle and it has no control over the memory buses
- The DMA controller takes over the buses to manage the transfer directly between the I/O devices and the memory unit



DMA (DIRECT MEMORY ACCESS)

- **Bus Request** It is used by the DMA controller to request the CPU to relinquish the control of the buses
- **Bus Grant** It is activated by the CPU to inform the external DMA controller that the buses are in high impedance state and the requesting DMA can take control of the buses Once the DMA has taken the control of the buses it transfers the data
- DMA transfer uses two signal
 - **HOLD**
 - **HLDA**

DMA (DIRECT MEMORY ACCESS)

➤ **HOLD**

- Active high input signal to 8085 from another master requesting the use of address and data bus
- After receiving the HOLD request, the MPU relinquishes the buses in the following machine cycle
- All buses are tri stated and HOLD acknowledge signal is sent out
- MPU regains the control of the buses after HOLD goes low

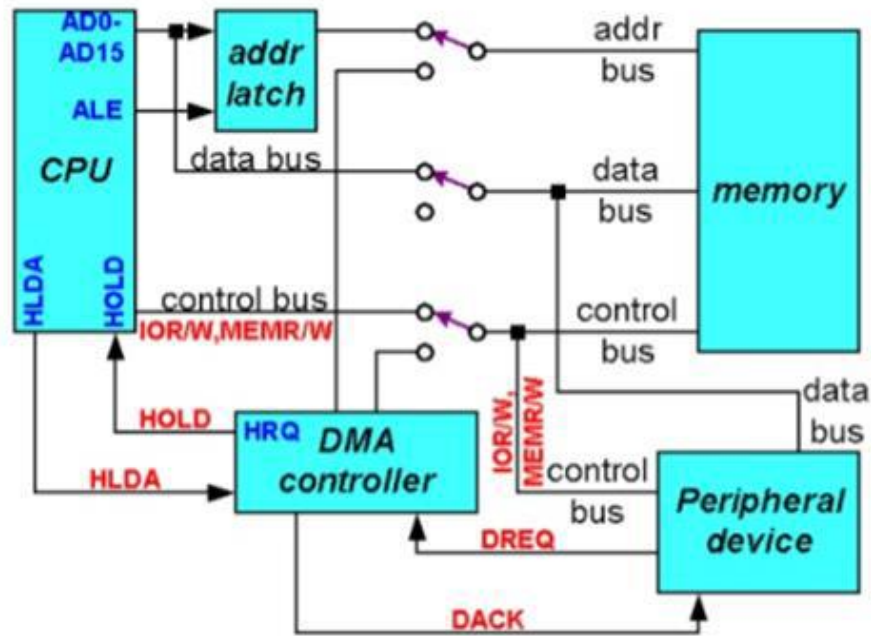
➤ **HLDA**

- This is an active high output signal indicting that MPU is relinquishing control of the buses
- A DMA controller uses these signals as if it were a peripheral requesting the MPU for the control of the buses

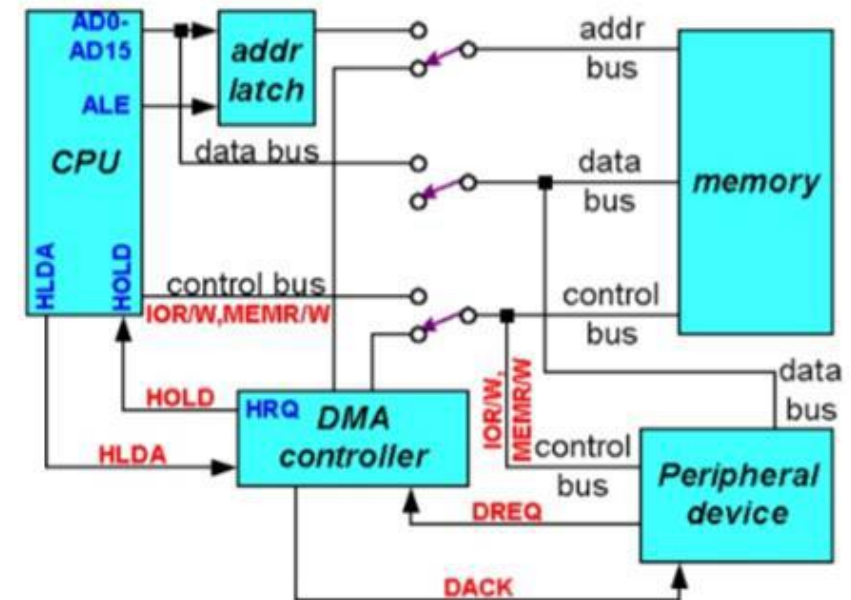
DMA (DIRECT MEMORY ACCESS)

The sequence of DMA Transfer

CPU having the control over the bus:



When DMA operates:



DMA (DIRECT MEMORY ACCESS)

The sequence of DMA Transfer

- Originally, microprocessor is connected to the memory as shown in fig with switches closed for address, data and control buses. When peripheral wants to transfer data using DMA transfer, it sends DMA request, DREQ, signal to the DMA controller.
- If the input (of the DMA controller is unmasked, the DMA controller will send a hold request, HRQ signal to the microprocessors HOLD input.
- The microprocessor finishes the current machine cycle and floats its buses, sending out a hold acknowledge signal, HLDA, to the DMA controller.

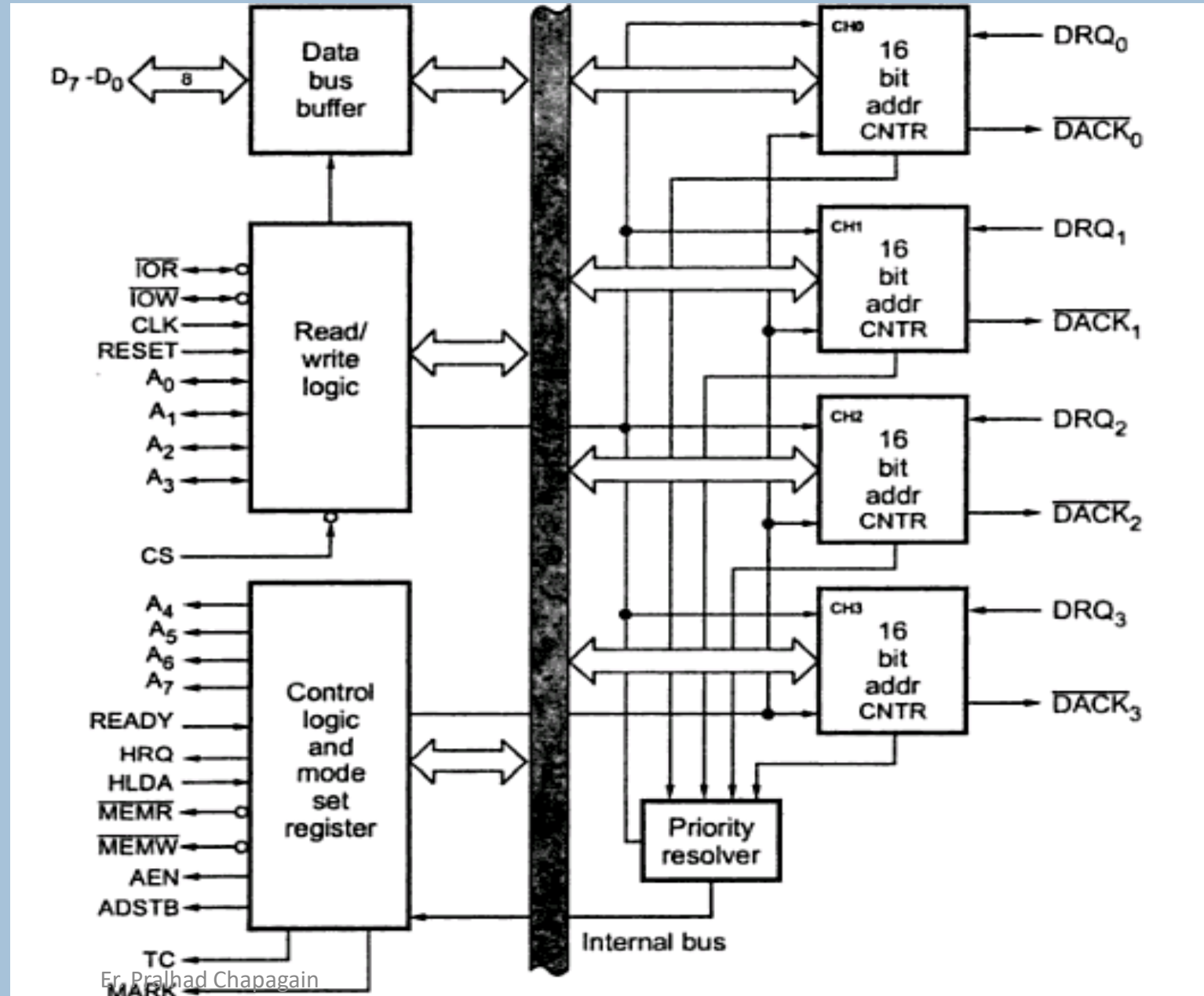
DMA (DIRECT MEMORY ACCESS)

- When DMA controller receives HLDA signal, it will send out a control signal which disconnects the processors from buses and connects DMA controller to the buses
Now DMA controller sends out the address of the byte to be transferred and send out DMA acknowledge (to the peripheral device to tell it to get ready to output the byte
- Then the DMA transfer begins and finally when the data transfer is complete, the DMA controller un asserts its hold request signal to the processor and releases the buses

DMA (DIRECT MEMORY ACCESS)

- DMA performs data transfer operation The different DMA transfer modes are as follows
 - Burst or Block transfer DMA
 - Cycle steal or Single byte transfer DMA
 - Transparent DMA

PROGRAMMABLE DMA CONTROLLER – INTEL 8257



PROGRAMMABLE DMA CONTROLLER – INTEL 8257

- It is a device to transfer the data directly between IO device and memory without through the CPU. So it performs a high-speed data transfer between memory and I/O device.
- The features of 8257 is,
 - The 8257 has four channels and so it can be used to provide DMA to four I/O devices.
 - Each channel can be independently programmable to transfer up to 64kb of data by DMA.
 - Each channel can be independently perform read transfer, write transfer and verify transfer.
- The functional blocks of 8257 as shown in the above figure are data bus buffer, read/write logic, control logic, priority resolver and four numbers of DMA channels.

PROGRAMMABLE DMA CONTROLLER – INTEL 8257

Operation of 8257 DMA Controller

- Each channel of 8257 has two programmable 16-bit registers named as address register and count register.
- Address register is used to store the starting address of memory location for DMA data transfer.
- The address in the address register is automatically incremented after every read/write/verify transfer.
- The count register is used to count the number of byte or word transferred by DMA.

PROGRAMMABLE DMA CONTROLLER – INTEL 8257

- In read transfer the data is transferred from memory to I/O device.
- In write transfer the data is transferred from I/O device to memory.
- Verification operations generate the DMA addresses without generating the DMA memory and I/O control signals.
- The 8257 has two eight bit registers called mode set register and status register.

Microprocessor System

1 Which of the following is the smallest entity of memory?

- (a) Block
- ☒ (b) Cell
- (c) Instance
- (d) Set

2 The primary memory (also called main memory) of a personal computer consists of

- (a) RAM only
- (b) ROM only
- ☒ (c) both RAM and ROM
- (d) Cache memory

3 The Boot sector files of the system are stored in which computer memory?

- (a) RAM
- ☒ (b) ROM
- (c) Cache
- (d) Register

4 Which of the following statements are not correct about the main memory of a computer?

- (a) In main memory, data gets lost when power is switched off.
- (b) Main memory is faster than secondary memory but slower than registers.
- (c) They are made up of semiconductors.
- ☒ (d) All are correct

5 Which of the following is the lowest in the computer memory hierarchy?

- (a) Cache
- (b) RAM
- ☒ (c) Secondary memory
- (d) CPU registers

6 Which of the following has the fastest speed in the computer memory hierarchy?

- (a) Cache
- ☒ (b) Register in CPU
- (c) Main memory
- (d) Disk cache

Microprocessor System

7 Which memory acts as a buffer between CPU and main memory?

- (a) RAM
- (b) ROM
- ☒ (c) Cache
- (d) Storage

8 Which of the following statements are not correct about cache memory?

- (a) Cache memory is used to store data temporarily.
- (b) It holds that data and program which has to be executed within a short period of time.
- (c) It consumes less access time as compared to the RAM.
- ☒ (d) All are correct.

9 Which computer memory chip allows simultaneous both read and write operations?

- (a) ROM
- ☒ (b) RAM
- (c) PROM
- (d) EEPROM

10 In which type of memory, once the program or data is written, it cannot be changed?

- (a) EPROM
- ☒ (b) PROM
- (c) EEPROM
- (d) None of these

11 In which type of ROM, data can be erased by ultraviolet light and then reprogrammed by the user or manufacturer?

- (a) PROM
- ☒ (b) EPROM
- (c) EEPROM
- (d) Both a and b

12 How many types of RAM are available?

- (a) 4
- (b) 3
- ☒ (c) 2
- (d) 5

Microprocessor System

13

What is true about memory unit?

- A. A memory unit is the collection of storage units or devices together.
- B. The memory unit stores the binary information in the form of bits.
- ☒ C. Both A and B
- D. None of the above

15

Which of the following is correct refreshed rate for DRAM?

- A. 10~1000 ms
- B. 10~50 ms
- ☒ C. 10~100 ms
- D. 10~500 ms

17

Suppose that a certain semiconductor memory chip has a capacity of $8K \times 8$. How many bytes could be stored in this device?

- a) 8,000
- b) 65,536
- ☒ c) 8,192
- d) 64,000

14

What is the formula for Hit Ratio?

- ☒ A. $\text{Hit}/(\text{Hit} + \text{Miss})$
- B. $\text{Miss}/(\text{Hit} + \text{Miss})$
- C. $(\text{Hit} + \text{Miss})/\text{Miss}$
- D. $(\text{Hit} + \text{Miss})/\text{Hit}$

16

The memory which is used to store the copy of data or instructions stored in larger memories, inside the CPU is called _____

- ☒ a) Level 1 cache
- b) Level 2 cache
- c) Registers
- d) TLB

18

What is the difference between static RAM and dynamic RAM?

- a) Static RAM must be refreshed, dynamic RAM does not
- b) There is no difference
- ☒ c) Dynamic RAM must be refreshed, static RAM does not
- d) SRAM is slower than DRAM

Microprocessor System

19

The device that enables the microprocessor to read data from the external devices is

- ☐ a) printer
- ☐ b) joystick
- ☐ c) display
- ☒ d) reader

20

The input and output operations are respectively similar to the operations,

- ☐ a) read, read
- ☐ b) write, write
- ☒ c) read, write
- ☐ d) write, read

21

The operation, IOWR (active low) performs

- ☐ a) write operation on input data
- ☒ b) write operation on output data
- ☐ c) read operation on input data
- ☐ d) read operation on output data

22

In memory-mapped scheme, the devices are viewed as

- ☐ a) distinct I/O devices
- ☒ b) memory locations
- ☐ c) only input devices
- ☐ d) only output devices

23

If a location is selected, then all the bits in it are accessible using a group of conductors called

- ☐ a) control bus
- ☐ b) address bus
- ☒ c) data bus
- ☐ d) either address bus or data bus

24

To address a memory location out of N memory locations, the number of address lines required is

- ☒ a) $\log N$ (to the base 2)
- ☐ b) $\log N$ (to the base 10)
- ☐ c) $\log N$ (to the base e)
- ☐ d) $\log (2N)$ (to the base e)

Microprocessor System

25 If the microprocessor has 10 address lines, then the number of memory locations it is able to address is

- a) 512
- ☒ b) 1024
- c) 2048
- d) none

27 Whenever a large memory is required in a microcomputer system, the memory subsystem is generally designed using

- a) Static RAM
- ☒ b) Dynamic RAM
- c) Both static and dynamic RAM
- d) ROM

29 The process of refreshing the data in the RAM to reduce the possibility of data loss is known as

- a) data cycle
- b) regain cycle
- c) retain cycle
- ☒ d) refresh cycle

26 To obtain 16-bit data bus width, the two 4K*8 chips of RAM and ROM are arranged in

- ☒ a) parallel
- b) serial
- c) both serial and parallel
- d) neither serial nor parallel

28 If a typical static RAM cell requires 6 transistors then corresponding dynamic RAM requires

- ☒ a) 1 transistor along with capacitance
- b) 2 transistors along with resistance
- c) 3 transistors along with diode
- d) 2 transistors along with capacitance

30 If 'n' denotes the number of rows that are to be refreshed in a single refresh interval, 'td' denotes the range of time it may take then, refresh time (t_r) can be defined as

- a) $n \cdot t_d$
- ☒ b) t_d / n
- c) n / t_d
- d) t_d^n

Microprocessor System

31

Port C of 8255 can function independently as

- a) input port
- b) output port
- c) either input or output ports
- d) both input and output ports

33

The data bus buffer is controlled by

- a) control word register
- b) read/write control logic
- c) data bus
- d) none of the mentioned

35

The device that receives or transmits data upon the execution of input or output instructions by the microprocessor is

- a) control word register
- b) read/write control logic
- c) 3-state bidirectional buffer
- d) none of the mentioned

32

All the functions of the ports of 8255 are achieved by programming the bits of an internal register called

- a) data bus control
- b) read logic control
- c) control word register
- d) none of the mentioned

34

The input provided by the microprocessor to the read/write control logic is

- a) RESET
- b) A1
- c) WR(ACTIVE LOW)
- d) All of the mentioned

36

If A1=0, A0=1 then the input read cycle is performed from

- a) port A to data bus
- b) port B to data bus
- c) port C to data bus
- d) CWR to data bus

Microprocessor System

37

37. The pin that clears the control word register of 8255 when enabled is

- a) CLEAR
- b) SET
- ☒ c) RESET
- d) CLK

38

38. How many pins does the 8255 PPI IC contains?

- a. 24
- b. 20
- c. 32
- ☒ d. 40

39

39. In which mode do all the Ports of the 8255 PPI work as Input-Output units for data transfer?

- a. BSR mode
- ☒ b. Mode 0 of I/O mode
- c. Mode 1 of I/O mode
- d. Mode 2 of I/O mode

41

41. In which of the following modes is the 8255 PPI capable of transferring data while handshaking with the interfaced device?

- a. BSR mode
- b. Mode 0 of I/O mode
- ☒ c. Mode 1 of I/O mode
- d. Mode 2 of I/O mode

40

40. Which of the following pins are responsible for handling the on the Read Write control logic unit of the 8255 PPI?

- a. CS'
- b. RD'
- c. WR'
- ☒ d. ALL of the above

42

42. How many bits of data can be transferred between the 8255 PPI and the interfaced device at a time? or What is the size of internal bus of the 8255 PPI?

- a. 16 bits
- b. 12 bits
- ☒ c. 8 bits
- d. None of the above

Microprocessor System

43

Q: Which port of the 8255 PPI is capable of performing the handshaking function with the interfaced devices?

- a. Port A
- b. Port B
- ☒ c. Port C
- d. All of the above

44

Q: In which of the following modes of the 8255 PPI, only port C is taken into consideration?

- ☒ a. BSR mode
- b. Mode 0 of I/O mode
- c. Mode 1 of I/O mode
- d. Mode 2 of I/O mode

45

Q: In mode 2 of I/O mode, which of the following ports are capable of transferring the data in both the directions?

- ☒ a. Port A
- b. Port B
- c. Port C
- d. All of the above

46

Q: In which of the following modes we do not consider the D6, D5 and D4 bits of the control word?

- ☒ a. BSR mode
- b. Mode 0 of I/O mode
- c. Mode 1 of I/O mode
- d. Mode 2 of I/O mode

47

Q: The 8255 ports works in the I/O mode,

- ☒ a. Programmable I/O ports
- b. Set pins
- c. Reset pins
- d. None of these

48

Q: Strobed input/output mode is also known as -

- a. Mode 0
- ☒ b. Mode 1
- c. Mode 2
- d. None of these

Microprocessor System

49 The serial communication is

- a) cheaper communication
- b) requires less number of conductors
- c) slow process of communication
- ☒ d) all of the mentioned

51 In _____ transmission of data, a group of n bits are sent simultaneously.

- ☒ parallel
- B. synchronous
- C. asynchronous
- D. serial

53 _____ transmission features start bits, stop bits, and gaps between data units

- A. Parallel
- B. Synchronous
- ☒ C. Asynchronous
- D. Virtual

50 The serial communication is used for

- a) short distance communication
- ☒ b) long distance communication
- c) short and long distance communication
- d) communication for a certain range of distance

52 Serial transmission can be _____

- A. parallel
- B. synchronous
- C. asynchronous
- ☒ D. b or c

54 Data transmission between the keyboard and the computer is usually _____

- A. parallel
- B. synchronous
- ☒ C. asynchronous
- D. virtual

Microprocessor System

55

Serial transmission without stop bits, start bits, or gaps is called _____ transmission

- A. parallel
- ☒ B. synchronous
- C. asynchronous
- D. virtual

56

In parallel transmission, if we are sending 80 Kbps (on each line), we send _____ characters per second

- ☒ A. 80,000
- B. 10,000
- C. 40,000
- D. none of above

57

The task of converting the byte into serial form and transmitting it bit by bit along with start, stop and parity bits is carried out by

- a) reception unit
- b) serial communication unit
- ☒ c) transmission unit
- d) all of the mentioned

58

Which of the following can be used for long distance communication?

- A. I²C
- B. Parallel port
- C. SPI
- ☒ D. RS232

59

Which of the following can provide hardware handshaking?

- ☒ A. RS232
- B. Parallel port
- C. Counter
- D. Timer

60

RS-232 is now known as _____

- A. OSI-232
- ☒ B. EIA-232
- C. ITU-232
- D. IEEE-232

Microprocessor System

61. A null modem has _____ connector(s).

- A. one male and one female
- B. two male
- ☒ C. two female
- D. only one

63. In the DB-25 implementation, the majority of the EIA-232 interface wires are used for _____

- A. data transmission
- ☒ B. control and timing
- C. test mode
- D. future assignment

65. The most popular implementation of EIA-232 is a _____-pin interface

- A. 23
- ☒ B. 25
- C. 232
- D. 9

62

In a null modem, the _____ pins are rewired so that DTEs can directly connect to each other

- A. data transfer
- B. set up
- C. timing
- ☒ D. all of the above

64

The _____ interface standard specifies a 4000-foot transmission distance

- A. EIA-232
- ☒ B. RS-422
- C. EIA-530
- D. X.21

66

The voltage level and the signal type are _____ specifications of the EIA-232 interface standard

- A. mechanical
- ☒ B. electrical
- C. functional
- D. any of the above

Microprocessor System

67 . The DMA transfers are performed by a control circuit called as _____

- a) Device interface
- ☒ b) DMA controller
- c) Data controller
- d) Overlooker

68 Which of the following provides an efficient method for transferring data from a peripheral to memory?

- ☒ A. dma controller
- ☐ B. serial port
- C. parallel port
- D. dual port

69 For transfer of data, devices inform the DMA through—?

- ☐ Bus request signal.
- ☒ DRQ.
- ☐ HLDA
- ☐ DAK

70 DMA acts as— to the CPU for data transfer.

- ☐ Master
- ☐ Slave.
- ☒ Both a and b.
- ☐ None of the above

71 In DMA transfers, the required signals and addresses are given by the _____

- a) Processor
- b) Device drivers
- ☒ c) DMA controllers
- d) The program itself

72 . After the completion of the DMA transfer, the processor is notified by _____

- a) Acknowledge signal
- ☒ b) Interrupt signal
- c) WMFC signal
- d) None of the mentioned

THANK YOU